
On the Structure of Reinforcement Learning Policies: Recovery from Synthetic Observations via Geometric Coverage

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Abstract

We study trained reinforcement learning policies as functions from observations to actions, asking whether a policy can be recovered from synthetic inputs that are independently sampled and often physically invalid. Across state-based MuJoCo locomotion tasks, teacher-labeled synthetic observations recover a large fraction of teacher performance without trajectories, rewards, or environment interaction. Recovery is governed by a two-factor prerequisite: synthetic samples must excite the directions along which the teacher varies, quantified by a sensitivity-overlap score $\text{Tr}(G_{\pi_T} \Sigma_q)$ that ranks controlled subspace conditions monotonically, and must provide minimal support across the full ambient input space. The second requirement is isolated by a thickening experiment: one-tenth of a standard deviation of orthogonal ambient noise raises recovery from 9.5% to 73.7% while barely changing the overlap score. Small amounts of real data localize the policy-relevant region; synthetic samples then provide coverage. Synthetic students inherit behavioral structure already encoded in the teacher, including perturbation responses and constraint-aware behavior, rather than acquiring new capabilities independently. These results support a *geometric coverage hypothesis*: exciting the teacher’s sensitive directions is equivalent to semantic trajectory-based supervision for behavioral cloning, with real data needed only to localize the policy-relevant region rather than to provide a qualitatively different supervisory signal.

1 Introduction

Reinforcement learning (RL) policies are usually studied through the procedures that produce them: exploration, long-horizon credit assignment, and repeated interaction with an environment [1, 2]. Yet the final object is simply a function from observations to actions. We ask whether this function is as complex as the training process that created it.

Standard imitation and policy distillation methods do not isolate this question. Behavioral cloning, DAgger, adversarial imitation, and policy distillation all learn from states drawn from environment trajectories or policy-induced distributions [3, 4, 5, 6, 7]. Offline imitation and offline RL similarly assume datasets of valid transitions or demonstrations [8, 9, 10, 11, 12]. Data-free distillation removes real data, but typically replaces it with optimized synthetic inputs designed to recover training-distribution information or elicit informative teacher responses [13, 14, 15, 16]. These methods implicitly optimize for high sensitivity-overlap by searching for inputs that maximize teacher activations or gradient norms. Our results suggest that for state-based policies, passive isotropic

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sampling already achieves substantial overlap due to the low effective dimension of the teacher’s sensitivity spectrum, and that the harder requirement is ambient support rather than directional search.

Recent work on subliminal learning shows that models can transmit latent behavioral traits through semantically unrelated data [17], with follow-up analyses connecting the effect to token-level and representation-level structure [18, 19]. These results suggest that learned behavior can persist in forms not visible from the ordinary semantics of data. We study an analogous question for RL policies in a more extreme setting: the inputs are not trajectories, not optimized synthetic examples, and not teacher-generated sequences. They are independently sampled synthetic observations, often physically invalid.

Surprisingly, this is often enough in state-based continuous-control policies. Across MuJoCo locomotion tasks, teacher-labeled synthetic observations recover a large fraction of teacher performance without trajectories or environment interaction. The effect is not explained by physical validity alone: shifted distributions fail, while uniform sampling can succeed in high-dimensional environments despite producing invalid states. This suggests that recovery depends less on semantic correctness or density matching, and more on whether samples satisfy two geometric requirements: covering input directions along which the policy’s action function varies, and providing minimal supervision across the full ambient observation space.

We present the Geometric Coverage Hypothesis: For state-based policies with low effective sensitivity dimension, the coverage component of trajectory-based supervision is geometrically equivalent to passive isotropic sampling. The residual gap between synthetic and semantic recovery is attributable entirely to the localization component, knowing where in the ambient space the policy-relevant region lies, which requires real observations and cannot be replaced by geometric coverage alone.

We formalize these requirements through a sensitivity-overlap framework. Trained policies exhibit anisotropic input sensitivity, with much of their action variation concentrated along a small set of sensitive directions. The sensitivity-overlap score $\text{Tr}(G_{\pi_T} \Sigma_q)$ quantifies how much a synthetic distribution excites these directions, and ranks recovery across controlled subspace interventions monotonically. A naive covering bound based on the effective dimension d_{sens} identifies the right geometric quantity but fails to capture either the empirical sample efficiency or the ambient support requirement. We propose a two-factor empirical prerequisite, $\mathcal{C}(q) \geq \tau$ for directional alignment and $\varepsilon \geq \varepsilon_{\text{min}}$ for ambient support, that is consistent with all controlled interventions in the paper and explains why full Gaussian sampling works reliably without directional engineering. A thickening experiment isolates the latter: strict subspace sampling collapses in closed-loop rollout despite optimal directional alignment, while a small amount of orthogonal ambient noise restores recovery. Mixing experiments further show that real data primarily provides localization, identifying where in normalized input space the teacher’s behavior matters, after which synthetic samples provide coverage. Lipschitz and gradient analyses confirm that smoothness alone is insufficient; successful recovery requires sensitivity aligned with the correct subspace, not merely low local complexity.

These findings connect to broader evidence that neural networks contain redundant functional structure [20], but our focus is distinct: we use synthetic probing to characterize what state-based RL policies look like as functions, not to propose a better imitation algorithm. We use synthetic distillation as a probe of policy structure. Our goal is not to compete with imitation learning when real data is available, nor to claim universality across all RL architectures or observation modalities, but to identify what synthetic supervision can and cannot recover from trained state-based policies.

Our contributions can be summarized as:

- **Passive policy recovery.** Teacher-labeled synthetic observations drawn from a simple isotropic distribution recover a large fraction of teacher performance without trajectories, rewards, or environment interaction. Controlled experiments rule out semantic explanations.
- **A geometric framework governing recovery.** Trained policies have anisotropic sensitivity concentrated along a small number of directions. The sensitivity-overlap score $\mathcal{C}(q) = \text{Tr}(G_{\pi_T} \Sigma_q)$ predicts recovery rank monotonically. Recovery additionally requires ambient support in orthogonal directions: without it, closed-loop trajectories leave the training subspace and performance collapses.
- **The Geometric Coverage Hypothesis.** Geometric coverage of the teacher’s excitation directions is equivalent to semantic trajectory-based coverage for behavioral cloning. The residual gap between synthetic and trajectory-based recovery is attributable entirely to

localization: real data identifies where in observation space the policy-relevant region lies; synthetic data covers it.

2 Synthetic Probing Setup

We study the deployed actor as a function from normalized policy inputs to actions. Let $\pi_T : \mathcal{X} \rightarrow \mathcal{A}$ denote a fixed teacher policy, where $\mathcal{X} \subseteq \mathbb{R}^d$ is the normalized input space consumed by the actor and $\mathcal{A} \subseteq \mathbb{R}^k$ is the action space. The teacher is trained by standard reinforcement learning and is held fixed throughout all probing experiments. We train a student policy $\pi_S : \mathcal{X} \rightarrow \mathcal{A}$ to match the teacher on synthetic inputs. The trained teacher policy is evaluated together with observation normalization statistics $(\mu_{\text{rms}}, \sigma_{\text{rms}})$ accumulated as running estimates during PPO training via the standard `VecNormalize` wrapper. These statistics are required only for executing the trained policy in the original environment: raw environment observations are normalized before being passed to the actor. Thus, `VecNormalize` metadata is part of the policy evaluation interface.

$(\mu_{\text{rms}}, \sigma_{\text{rms}})$ are not used to sample, center, scale, fit, or otherwise construct the synthetic training inputs. Instead, all synthetic distributions are specified directly in the normalized policy-input coordinate system. The baseline distribution is $x \sim \mathcal{N}(0, I_d)$, and we also consider Laplace, uniform, narrow, wide, and shifted variants in the same normalized coordinate system. These distributions are fixed simple distributions over actor inputs; they are not estimated from rollout states, do not use empirical state covariances, and are not chosen to approximate the environment state distribution. In all cases, synthetic inputs are sampled independently. They are not produced by environment rollouts, do not respect transition dynamics, and need not correspond to physically valid raw states.

This distinction matters because normalized rollout observations are not actually distributed as $\mathcal{N}(0, I_d)$. The stored normalization statistics are running estimates accumulated during training, while the policy-induced state distribution changes as the policy improves. Consequently, real rollout inputs in normalized coordinates remain correlated, only approximately centered, and generally non-Gaussian. The baseline $\mathcal{N}(0, I_d)$ distribution is therefore a simple coordinate-system probe, not a Gaussian fitted to the teacher’s state distribution. Given a synthetic dataset $\mathcal{D}_{\text{syn}} = \{(x_i, a_i)\}_{i=1}^N$, $a_i = \pi_T(x_i)$, where x_i denotes a normalized policy-input vector, the student minimizes $\mathcal{L}(\pi_S) = N^{-1} \sum_i \|\pi_S(x_i) - \pi_T(x_i)\|_2^2$. No environment interaction occurs during student training. The teacher is queried only on independently sampled normalized inputs, and the synthetic inputs are not optimized to maximize activations, match internal statistics, or approximate the empirical state distribution.

We evaluate the trained student by executing it in the original environment under nominal dynamics. During evaluation, raw environment observations are normalized using the stored `VecNormalize` statistics before being passed to the student, exactly as for the teacher. Let $R(\pi)$ denote the mean episodic return of policy π over 100 evaluation rollouts. We report recovery as percentage of teacher performance: $\text{Recovery}(\pi_S) = 100 \times \frac{R(\pi_S)}{R(\pi_T)}$. Values above 100% are possible because the teacher is not necessarily the unique maximizer of return under deterministic evaluation.

This setup is intentionally restrictive. It removes the three sources of structure usually available in imitation or distillation: valid trajectories, temporal correlations, and input distributions chosen to resemble the environment. If a student trained in this way recovers the teacher’s behavior, the recovery must arise from structure in the teacher policy as a function over normalized actor inputs, rather than from physical validity or semantic correctness of the inputs.

3 Recovery from Synthetic Observations

We first ask whether synthetic observations alone contain enough information to recover a trained policy. The student is trained only on independently sampled teacher-labeled inputs from simple distributions. No real trajectories, temporal structure, rewards, or environment interaction are used during student training.

Synthetic observations recover substantial teacher performance. Figure 1 shows that synthetic supervision recovers a large fraction of teacher performance across four MuJoCo locomotion tasks. Hopper-v5 recovers near-teacher performance using 10^5 synthetic samples, Walker2d-v5 reaches teacher-level recovery by 5×10^5 samples, and Ant-v5 approaches teacher performance at 2×10^6

samples. Humanoid-v5 is harder but still recovers a substantial fraction of teacher performance despite its 348-dimensional observation space. Full numerical results are reported in Appendix D.2.

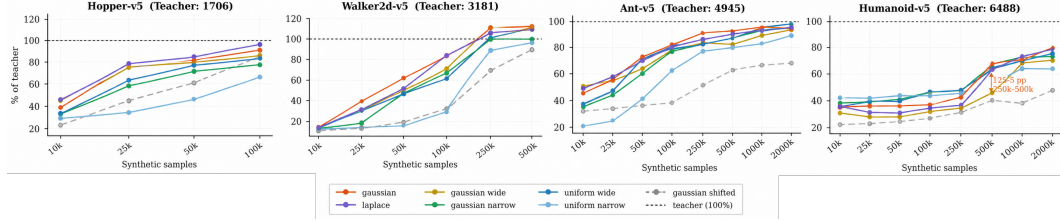


Figure 1: Synthetic recovery across MuJoCo environments. Recovery improves with sample count. The shifted Gaussian degrades performance relative to centered baselines, while uniform sampling becomes competitive in higher-dimensional tasks.

The positive intuition is geometric rather than semantic. Synthetic inputs need not be valid states if they excite directions along which the teacher’s action function varies and provide enough ambient support for closed-loop deployment. Section 4 formalizes this through sensitivity overlap and subspace interventions.

Recovery is not explained by physical validity alone. A natural explanation is that synthetic samples work only because they approximate the empirical state distribution. Figure 1 challenges this view: centered Gaussian and Laplace distributions perform well, and uniform sampling becomes competitive in Ant-v5 and Humanoid-v5 despite producing physically invalid inputs. Thus, semantic or physical validity is not required for useful supervision. The shifted distribution provides a complementary stress test. It can fail badly even when its coordinate-wise scale is comparable to successful centered distributions, showing that arbitrary synthetic distributions do not work. However, it is not a clean geometric control: in high-dimensional normalized space, shifting every coordinate by a constant greatly increases aggregate displacement and can push tanh networks toward saturation. We therefore do not use shifted-distribution failure alone as evidence against Euclidean proximity.

The stronger evidence comes from controls developed below. Uniform-wide sampling succeeds in high dimensions despite physical invalidity, with a concentration-of-measure explanation in Appendix D.13. The real-fit Gaussian diagnostic in Appendix D.10 shows that, in Humanoid-v5, covariance structure matters far more than recentering. Most directly, the subspace intervention in Section 4.3 holds dimension and per-direction variance fixed while changing only the sampled eigenspace; teacher-sensitive directions recover far more behavior than random or insensitive subspaces.

Table 14 further addresses the shifted-baseline confound. A norm-controlled random shift with $c = 0.75$ still underperforms the centered Gaussian baseline, showing that the shifted-distribution gap is not solely an artifact of increased input norm or activation saturation. **Scaling reveals a coverage threshold.** The same figure also reveals a scaling pattern. Hopper-v5 saturates at moderate sample counts; Walker2d-v5 improves steadily to teacher-level recovery by 5×10^5 samples; Ant-v5 and Humanoid-v5 require larger budgets. Humanoid-v5 shows the clearest threshold behavior, rising sharply between 2.5×10^5 and 5×10^5 samples. This suggests that recovery begins once the synthetic dataset sufficiently covers policy-relevant directions.

Implication. The recovery results rule out the simplest interpretations: the student is not learning from trajectories, rewards, environment interaction, or physically valid states. The remaining explanation is geometric: recovery depends on covering the directions and regions of input space that control the teacher’s action function.

The next section develops this interpretation through a sensitivity-overlap framework, showing that directional alignment and ambient support jointly govern recovery. Smoothness alone, however, does not predict recovery success (Section 4.6).

4 Geometry of Policy Recovery

The previous section establishes the empirical puzzle: synthetic observations can recover high-performing policies even when they are physically invalid, while some distributions fail despite

comparable coordinate scale. We now give a geometric interpretation. Recovery requires two conditions: synthetic samples must excite directions along which the teacher varies, and they must provide minimal support across the full ambient input space. We use *policy-relevant directions* for the top eigendirections of the empirical sensitivity matrix G_π , and *ambient support* for nonzero variance in directions orthogonal to the dominant sensitivity subspace.

4.1 Sensitivity overlap

We formalize directional excitation using the teacher sensitivity matrix

$$G_{\pi_T} = \mathbb{E}_{x \sim \rho_{\text{eval}}} [J_{\pi_T}(x)^\top J_{\pi_T}(x)], \quad (1)$$

where $J_{\pi_T}(x)$ is the teacher Jacobian on real rollout observations. For a zero-mean synthetic distribution q with covariance Σ_q , a first-order approximation gives the sensitivity-overlap score

$$\mathcal{C}(q) = \text{Tr}(G_{\pi_T} \Sigma_q). \quad (2)$$

Intuitively, if the teacher is sensitive along direction v_j , then q excites that direction with strength $v_j^\top \Sigma_q v_j$; Eq. 2 sums these excitations over teacher-sensitive directions, weighted by the corresponding eigenvalues. If $G_{\pi_T} = U \Lambda U^\top$ with eigenvalues $\lambda_1 \geq \dots \geq \lambda_d$, isotropic sampling in the top- k eigenspace gives $\sum_{j=1}^k \lambda_j$, a random k -subspace gives expected overlap $\frac{k}{d} \sum_j \lambda_j$, and the bottom- k eigenspace gives $\sum_{j=d-k+1}^d \lambda_j$. Thus, at matched dimension and variance, teacher-sensitive subspaces should provide more useful supervision than random or insensitive ones.

Why a naive covering bound fails. A standard ε -net covering argument suggests that $N \gtrsim (LD/\varepsilon)^{d_{\text{sens}}}$ samples would be needed to approximate an L -Lipschitz function to error ε over a region of effective dimension d_{sens} . This identifies d_{sens} as relevant, but misses two empirical facts. First, recovery is better organized by $\mathcal{C}(q)$ than by covering radius. Second, closed-loop recovery also requires ambient support: strict Top-78 sampling fails at 9.5% recovery despite near-maximal overlap, while adding $\varepsilon = 0.10$ of orthogonal noise restores recovery to 73.7%. This ambient requirement arises from deployment in the full d -dimensional observation space.

A two-factor heuristic for recovery. The experiments suggest an empirical prerequisite: $\mathcal{C}(q) \geq \tau$ and $\varepsilon \geq \varepsilon_{\min}$, where ε denotes the scale of ambient support orthogonal to the dominant sensitivity subspace. This is not a finite-sample return bound, but a predictive decomposition: directional alignment determines which solutions are reachable, while ambient support determines whether they remain stable under closed-loop rollout.

4.2 Sensitivity spectra and synthetic probing

We estimate G_π as in Eq. 1 on real rollout observations and report r_{90} , the number of directions explaining 90% of total sensitivity, and the participation-ratio dimension $d_{\text{sens}} = \frac{(\sum_j \lambda_j)^2}{\sum_j \lambda_j^2}$.

Table 1: Sensitivity structure under real and synthetic inputs on Humanoid-v5 ($d_{\text{obs}} = 348$, 10 seeds). $\cos \bar{\theta}$ measures alignment with the real-observation teacher eigenspace at the dominant top- k subspace ($k = 16$) and the broader r_{90} subspace ($k = 78$). Full extensions are in Appendix D.7.

Source	d_{sens}	$\cos \bar{\theta}$ (top- k)	$\cos \bar{\theta}$ (r_{90})
Real	15.6 ± 1.2	1.000	1.000
Gaussian	17.5 ± 0.8	0.988 ± 0.003	0.764 ± 0.034
Uniform	12.7 ± 0.6	0.988 ± 0.003	0.766 ± 0.032
Scale mixture	15.0 ± 0.7	0.988 ± 0.003	0.764 ± 0.032
Gaussian-unif. mix.	14.0 ± 0.6	0.988 ± 0.002	0.767 ± 0.033
Shifted	21.5 ± 1.1	0.949 ± 0.007	0.724 ± 0.033
Norm-shift ($c = 0.75$)	15.7 ± 0.7	0.955 ± 0.005	0.729 ± 0.030

Table 1 shows a two-level structure. At the dominant scale ($k = 16$), centered synthetic distributions recover the teacher eigenspace almost identically ($\cos \bar{\theta} \approx 0.988$), while shifted distributions align less well (0.949–0.955). This explains why simple centered distributions can succeed despite not

resembling valid states: they excite the same dominant teacher-sensitive directions. At the broader r_{90} scale, all centered isotropic distributions converge to a common floor ($\cos \bar{\theta} \approx 0.76$), suggesting that distributional shape alone cannot recover the sensitivity tail. The real-fit Gaussian analysis in Appendix D.10 shows that off-diagonal rollout covariance accounts for much of the remaining gap.

4.3 Directional coverage through subspace intervention

Sensitivity spectra are correlational. We therefore sample in controlled subspaces: let $V_k \in \mathbb{R}^{d \times k}$ be an orthonormal basis, sample $z \sim \mathcal{N}(0, I_k)$, and set $x = V_k z$. This fixes per-active-direction variance and changes only the support directions. At budget 10^5 and 20 epochs, we compare Full Gaussian, Top- k , Random- k , and Bottom- k for $k = \lceil d_{\text{sens}} \rceil$ and $k = r_{90}$.

Table 2: Subspace intervention recovery on Humanoid-v5 and Ant-v5 (mean $\pm$ std, 10 seeds). Top- k uses the top eigendirections of G_{π_T} , Random- k uses a randomly sampled k -dimensional subspace, and Bottom- k uses the least-sensitive eigendirections. Normalized sensitivity-overlap scores are shown in parentheses.

Condition	Humanoid (%)	Ant (%)
Full Gaussian (1.000)	39.8 ± 18.0	81.0 ± 11.2
Top- d_{sens} (0.810 / 0.736)	16.4 ± 9.9	25.8 ± 13.1
Random- d_{sens} (0.046 / 0.067)	3.6 ± 1.8	18.7 ± 2.5
Bottom- d_{sens} (0.003 / 0.002)	2.8 ± 0.8	16.9 ± 4.3
Top- r_{90} (0.901 / 0.897)	25.0 ± 9.1	39.6 ± 19.3
Random- r_{90} (0.224 / 0.162)	4.5 ± 1.3	16.5 ± 3.7

The subspace intervention confirms the directional hypothesis. In Humanoid, Top-16 recovers 16.4%, compared with 3.6% for Random-16 and 2.8% for Bottom-16. Random-16 is not exactly zero because any low-dimensional subspace captures some sensitivity by chance, but its expected overlap is only 0.046. The gap between Top-16 and Full Gaussian also shows why directional overlap alone is insufficient: even with high overlap, a strict 16-dimensional subspace provides no supervision in the remaining ambient directions, so the student must extrapolate during closed-loop rollout. Top-78 captures roughly 90% of teacher sensitivity and improves to 25.0%, while Random-78 remains at 4.5%. Across conditions, the overlap score ranks recovery monotonically on Humanoid (Spearman $\rho = 1.00$) and strongly on Ant ($\rho = 0.83$).

4.4 Ambient support requirement

Directional overlap alone is insufficient. Top-78 has overlap 0.901, close to Full Gaussian at 1.000, but recovers only 25.0% versus 39.8%. The gap arises because a student trained strictly inside a subspace receives no supervision off that subspace, yet closed-loop rollouts need not remain exactly inside it. We isolate this with a thickening experiment:

$$x = V_{78} z_{\top} + \varepsilon V_{\perp} z_{\perp}, \quad z_{\top} \sim \mathcal{N}(0, I_{78}), \quad z_{\perp} \sim \mathcal{N}(0, I_{d-78}). \quad (3)$$

Table 3: Thickened Top-78 recovery on Humanoid-v5. Values are mean $\pm$ std across 10 seeds at 500k samples and 100 epochs. $\varepsilon = 0$ is strict Top-78; $\varepsilon = 1$ approximates Full Gaussian.

ε	Overlap / $\text{Tr}(G)$	Recovery (%)
0	0.901 ± 0.014	9.5 ± 2.7
0.05	0.899 ± 0.015	56.8 ± 19.4
0.10	0.900 ± 0.015	73.7 ± 19.7
0.25	0.905 ± 0.014	71.7 ± 14.0
0.50	0.924 ± 0.011	73.3 ± 17.4
1.00	1.000 ± 0.000	63.5 ± 19.3

Adding orthogonal noise triggers a sharp phase transition (Table 3). Strict Top-78 recovers only 9.5%. Adding $\varepsilon = 0.10$ of orthogonal noise barely changes overlap ($0.901 \rightarrow 0.900$) but raises recovery

to 73.7%. Recovery remains stable for $\varepsilon \in \{0.10, 0.25, 0.50\}$ and drops only at $\varepsilon = 1.00$, where directional concentration is lost. Thus, sensitivity overlap explains why top eigendirections beat random or bottom eigendirections, while ambient support explains why strict subspace training fails in rollout.

The decrease at $\varepsilon = 1.00$ is also informative. At this endpoint the sampling distribution regains full ambient support, but no longer concentrates probability mass in the teacher-sensitive subspace. Performance therefore drops relative to the lightly thickened conditions even though the overlap score is maximal by normalization. The best recovery occurs in the intermediate regime: enough orthogonal variance to constrain off-subspace behavior, but enough concentration along the top eigendirections to preserve directional signal. This is the empirical form of the two-factor prerequisite.

4.5 Students recover the teacher’s sensitive coordinates

We next test whether students recover the teacher’s actual sensitivity structure. For each policy, we compute coordinate-wise sensitivity and compare teacher and student using Spearman correlation, cosine similarity, and top- k eigenspace overlap.

Table 4: Teacher-student sensitivity $I_j(\pi) = \frac{1}{n} \sum_i \|J_i[:, j]\|_2^2$ alignment on Humanoid-v5 ($k = \lceil d_{\text{sens}} \rceil = 16$).

Distribution	Samples	Spearman ρ	Cosine sim.	Subspace $\cos \bar{\theta}$
Gaussian shifted	25k	0.939	0.991	0.947
Gaussian baseline	25k	0.981	0.996	0.986
Gaussian narrow/wide	25k	0.978–0.983	0.995–0.997	0.984–0.987
All distributions	500k	0.989–0.992	0.996–0.998	≥ 0.985
All distributions	2M	0.993–0.995	0.998–0.999	≥ 0.985

Table 4 shows high teacher-student alignment even at 25k samples, tightening with scale to Spearman correlation near 0.995 and subspace overlap $\cos \bar{\theta} \geq 0.985$. Thus, students do not learn an unrelated low-dimensional approximation; they recover the teacher’s dominant sensitivity structure. Per-environment extensions are in Appendices D.8 and D.4.

4.6 Smoothness is not sufficient

Finally, we measure an empirical Lipschitz estimate, mean action-logit magnitude, and input-gradient norm on real observations. Students preserve action scale while matching or reducing teacher sensitivity. However, smoothness alone does not explain recovery. On Humanoid-v5, shifted students are smoother than Gaussian-baseline students (Lipschitz 0.203 vs. 0.257; gradient 3.20 vs. 3.87×10^{-5}) but recover worse (22.5% vs. 36.6%; Appendix D.6). Successful recovery requires local sensitivity aligned with the correct subspace, not merely lower local complexity.

Table 5: Teacher and synthetic student geometry on real environment observations. Student values are averaged over synthetic configurations.

Environment	Teacher Lip.	Student Lip.	Teacher logit	Student logit	Teacher grad.	Student grad.
Ant-v5 (10^{-5})	0.19 ± 0.01	0.16 ± 0.01	0.94 ± 0.07	0.90 ± 0.06	2.48 ± 0.12	2.12 ± 0.14
Humanoid-v5 (10^{-5})	0.25 ± 0.01	0.25 ± 0.01	3.03 ± 0.18	3.24 ± 0.14	4.07 ± 0.27	3.88 ± 0.14
Hopper-v5 (10^{-4})	2.71 ± 2.24	0.79 ± 0.09	1.69 ± 0.56	1.55 ± 0.07	7.77 ± 7.02	2.09 ± 0.26
Walker2d-v5 (10^{-4})	0.85 ± 0.15	0.85 ± 0.15	3.08 ± 0.48	3.08 ± 0.48	1.14 ± 0.20	1.14 ± 0.20

Together, these results support the two-factor view. Trained policies are anisotropic; synthetic inputs recover the teacher’s dominant sensitivity subspace; directional alignment predicts recovery rank; and ambient support is separately necessary for closed-loop success. The next section asks why, if synthetic samples provide both, small real buffers still improve recovery in harder environments.

5 Mixed Real-Synthetic Recovery in the Low-Data Regime

The fully synthetic setting isolates recoverable structure in the teacher’s action function but scales poorly to high-dimensional observation spaces. We therefore study a low-data regime where the teacher can be queried, a small buffer of real rollout observations is available, and no additional environment interaction is permitted. Our goal is to test whether real data primarily *localizes* the policy-relevant region, meaning it identifies where in normalized input space the teacher’s behavior matters, while synthetic samples provide *coverage*.

Students are trained on a fixed budget of 100k teacher-labeled inputs, partitioned between real rollout observations and synthetic samples. All labels come from the teacher under the same supervised regression objective, ensuring that real data contributes spatial localization rather than a supervision signal. To disentangle localization from temporal structure, we compare two real-data variants: Variant A uses contiguous rollout segments (preserving sequence order), while Variant B draws i.i.d. samples from a rollout pool (preserving the state marginal but removing temporal correlations). Behavioral cloning and DAGger [4] trained on real data at matched budgets serve as baselines.

This fixed-budget design should not be read as a claim that synthetic labels are preferable to fully real data when large real buffers are available. The target regime is more constrained: a practitioner has a small real buffer, can query a teacher policy offline, but cannot collect additional environment data. The relevant question is therefore whether synthetic teacher queries can amplify a limited real anchor, and whether the gain comes from the marginal location of visited states or from trajectory order.

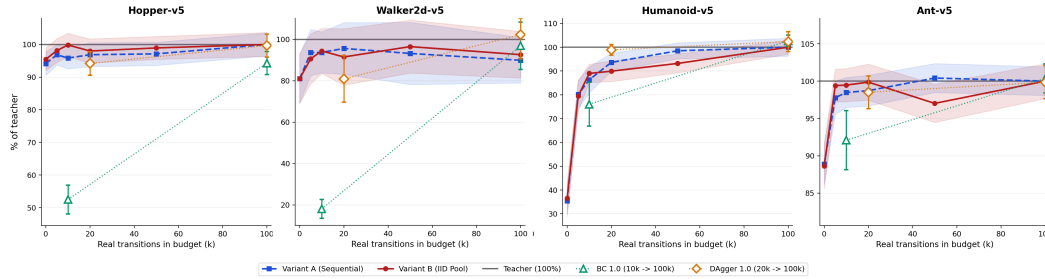


Figure 2: Fixed-budget real-synthetic mixing. Variant A uses sequential rollout observations; Variant B samples iid observations from a rollout pool. BC and DAGger show fully real-data baselines at comparable budgets.

Figure 2 shows that real data provides sharp, saturating gains in high-dimensional environments but minimal gains where synthetic recovery is already near-teacher. In Humanoid-v5 and Walker2d-v5, adding only 5k–20k real observations raises performance sharply; beyond this anchor, additional real data yields diminishing returns. In Hopper-v5 and Ant-v5, pure synthetic recovery is already high, indicating the isotropic distribution already covers the relevant region. The marginal value of real data is thus largest where synthetic localization is hardest.

These results support a clean decomposition:

$$\text{real data} \rightarrow \text{localization}, \quad \text{synthetic data} \rightarrow \text{coverage}.$$

Real observations identify where in normalized input space the teacher’s behavior matters; synthetic queries then densely cover that region.

This interpretation also explains why the mixed curves saturate quickly. Once a small real buffer anchors the relevant region, additional real observations are partly redundant under the fixed 100k-label budget: they improve localization, but they also displace synthetic queries that would otherwise broaden coverage. Conversely, when the synthetic distribution already covers the relevant region well, as in Hopper-v5 and Ant-v5, adding real samples has little marginal value. The useful role of real data is therefore not to provide a different target, since all labels are still teacher actions, but to select the part of normalized input space where teacher matching matters for rollout performance.

The comparison is intentionally different from standard behavioral cloning. A fully real 100k-observation dataset remains a strong reference because it supplies both localization and coverage from the rollout distribution. The mixed setting asks a more constrained question: if only a small

real buffer is available, can synthetic teacher queries supply the missing coverage without new environment interaction? The answer is environment-dependent but positive in the hard cases: small real anchors close much of the gap in Walker2d-v5 and Humanoid-v5, while fully synthetic recovery already suffices in easier settings.

Trajectory structure is not the main source of the gain. Comparing contiguous rollout segments (Variant A) against i.i.d. samples from the same rollout pool (Variant B) isolates the role of temporal ordering. The two variants yield statistically indistinguishable recovery across all environments and budgets (paired tests in Appendix D.12), confirming that the mixed-data benefit stems from marginal state coverage rather than trajectory structure. Consecutive transitions are not required.

Implication. Fully synthetic distillation is a geometric probe; mixed recovery is a low-data strategy. When interaction is unavailable, a small real buffer anchors the relevant region and synthetic teacher queries amplify it.

6 Discussion and Conclusion

We studied trained reinforcement learning policies as functions from observations to actions. The central finding is that state-based continuous-control policies can be recovered from independently sampled synthetic inputs that are not valid environment states and contain no temporal or reward information: the useful signal is not semantic correctness, but the geometry of how the teacher maps inputs to actions.

Recovery requires two geometric conditions. Synthetic samples must cover policy-sensitive directions, quantified by the sensitivity-overlap score $\text{Tr}(G_{\pi_T} \Sigma_q)$, which ranks recovery conditions monotonically on Humanoid-v5 (Spearman $\rho = 1.00$). They must also provide minimal support across the full ambient observation space. The second condition is not captured by the sensitivity spectrum alone: the thickening experiment isolates it directly. A student trained within the teacher’s top-78 sensitive directions fails at 9.5% recovery despite near-maximal overlap, while $\varepsilon = 0.10$ of orthogonal ambient noise restores recovery to 73.7%. Full Gaussian sampling satisfies both conditions simultaneously by construction, which is why it works reliably without directional engineering.

These findings support a *geometric coverage hypothesis*: exciting the teacher’s sensitive directions is equivalent to semantic trajectory-based supervision for behavioral cloning. The residual gap between purely synthetic and trajectory-based recovery is attributable not to a qualitatively different supervisory signal, but to localization — knowing where in the ambient observation space the policy-relevant region lies. The mixed-data experiments make this decomposition precise. Real data primarily provides localization: it supplies the off-diagonal covariance structure that isotropic synthetic marginals cannot, identifying where in normalized input space the teacher’s behavior matters. Synthetic samples then provide coverage once the relevant region is identified. This explains why small real buffers sharply improve recovery in harder environments while synthetic samples can replace much of the real data in easier ones. The practical implication is not that real data can be discarded, but that small real buffers can be amplified by synthetic teacher queries when new environment interaction is unavailable.

It is important not to interpret the standard Gaussian baseline as matching the teacher’s rollout distribution. Although VecNormalize approximately centers and scales observations, its running statistics lag the current policy-induced state distribution. Normalized rollout observations remain correlated, only approximately centered, and not distributed as $\mathcal{N}(0, I)$. Recovery from $x \sim \mathcal{N}(0, I)$ is therefore not simply recovery from the training distribution; it is recovery from a distribution that satisfies the two geometric requirements but does not match the rollout marginal. The useful property of the Gaussian baseline is its geometry, not its proximity to the empirical state distribution.

The perturbation (B) and SafetyPointGoal1 (C) results further clarify the scope of the claim. Synthetic distillation does not create robustness, safety, or dynamics knowledge independently. The student receives no rewards, costs, constraints, transitions, or robustness objective. It inherits whatever behavioral structure is already encoded in the teacher’s observation-to-action map, including strengths, failures, perturbation responses, and constraint-aware behavior.

Overall, synthetic probing reveals that trained state-based RL policies are not arbitrary black-box mappings over the full observation space. They behave like structured, anisotropic functions whose task-relevant behavior is concentrated along a small set of sensitive directions, with the remaining

directions requiring only minimal ambient supervision for closed-loop viability. This perspective does not replace standard imitation learning when sufficient real data is available, but it separates what real data contributes, what synthetic coverage can recover, and which behavioral properties are encoded directly in trained policy functions.

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A Limitations and Scope

Our empirical scope is intentionally focused. The main experiments study compact vector-observation continuous-control policies, primarily PPO-family teachers in MuJoCo locomotion. SafetyPointGoal1 extends the objective class to constrained reward-cost training with PPO-Lag, but we do not claim algorithm-universal recovery. Off-policy stochastic actors such as SAC, discrete-action value-based policies, recurrent or Transformer policies, graph-structured controllers, multi-agent policies, and visual policies require separate study because they change either the policy input structure, the action representation, or the relevant notion of teacher matching.

The phenomenon studied here appears to depend on policies receiving compact vector observations. We do not find that the same procedure transfers directly to pixel-based policies. This is consistent with the role of architecture: convolutional networks rely on spatially organized inputs so that local filters and hierarchical activations fire meaningfully. Independent noise does not preserve this spatial structure and therefore does not provide the same kind of useful teacher-query signal. Although prior work on subliminal learning reports preliminary image-domain results on MNIST, those experiments use fully connected networks rather than convolutional policies [17]. We were able to replicate the fully connected MNIST setting, but the effect did not carry over to Atari or other pixel-based RL environments. Thus, our claims are restricted to state-based policies and should not be read as evidence that arbitrary noise can recover visual policies.

The real-fit Gaussian diagnostic in Appendix D.10 also clarifies the cost of being data-free. Sampling from a Gaussian fitted to teacher rollouts substantially closes the recovery gap on Humanoid-v5 and Ant-v5, with off-diagonal covariance structure rather than mean centering accounting for most of the improvement. This baseline is real-data-informed and does not displace the central result; instead, it sharpens the interpretation. High-dimensional recovery benefits from sampling distributions that respect the correlated geometry of the policy-relevant region.

Finally, synthetic distillation should not be interpreted as a universal policy-improvement method. Synthetic queries can provide coverage of a teacher’s observation-to-action map without additional environment interaction, but they do not automatically localize the region where that map matters. The most plausible algorithmic uses are teacher compression, policy probing, behavioral auditing, and synthetic augmentation of small real buffers or offline datasets. In high-dimensional settings, synthetic augmentation should be combined with real-state anchors, rollout-fitted structure, or other mechanisms that localize sampling to the policy-relevant region.

Impacts. Synthetic policy probing could enable behavioral auditing of trained policies without environment access, which has safety implications for deployed RL systems. Students inherit teacher behavioral structure including failures, which means distillation pipelines can propagate unintended behaviors. No direct negative applications are anticipated from this work.

B Behavioral Fidelity Under Perturbations and Constraints

The previous sections show that synthetic observations can recover the teacher’s policy structure under nominal dynamics. We now ask whether students also preserve the teacher’s response under

off-nominal evaluation. This is not a test of whether synthetic distillation creates robustness beyond the teacher: the student receives no reward signal, environment interaction, domain randomization, or robustness objective. The relevant question is whether the student tracks the teacher’s perturbation-response profile.

We evaluate teacher, behavioral cloning, DAgger, and synthetic students under additive observation noise, gravity scaling, and body-mass scaling. Policies are evaluated without retraining, and returns are normalized by each method’s own unperturbed return.

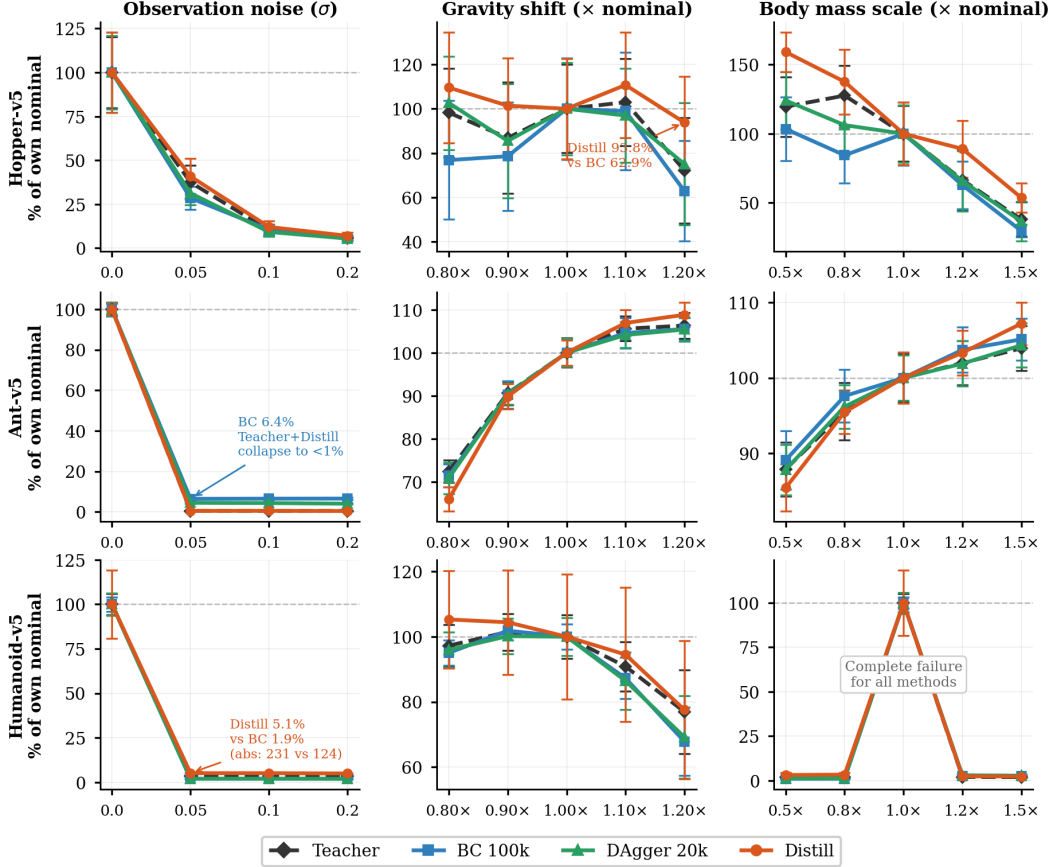


Figure 3: Perturbation fidelity under observation and dynamics shifts. Returns are normalized by each method’s nominal return. Synthetic students generally inherit the teacher’s perturbation profile: they degrade when the teacher degrades and often retain performance when the teacher does. The goal is behavioral fidelity, not robustness improvement.

Figure 3 shows that synthetic students largely inherit the teacher’s perturbation profile. Observation noise gives the clearest case: across Hopper-v5, Humanoid-v5, and Ant-v5, the teacher degrades sharply as noise increases, and synthetic students degrade in the same regime. This is expected. The student has no mechanism for learning invariance to arbitrary observation corruption unless such invariance is already encoded in the teacher’s action map and covered by the distillation distribution.

Dynamics perturbations are more environment-dependent. In Hopper-v5, synthetic students retain more nominal-normalized performance than behavioral cloning under several gravity and mass shifts. In Humanoid-v5, gravity shifts degrade all methods similarly, and body-mass scaling causes near-complete failure except near nominal mass. Ant-v5 remains comparatively stable under gravity and mass shifts but collapses under observation noise.

These results separate policy-function fidelity from robustness improvement. Synthetic recovery can match the teacher’s observation-to-action behavior well enough that downstream rollout behavior co-varies with the teacher under perturbation. It does not learn new invariances or control strategies absent from the teacher. In this sense, the student inherits the teacher’s behavioral geometry, including

both strengths and failures. Appendix C gives the complementary constrained-control case: when the teacher is trained with a reward-cost objective, synthetic students can also inherit safety-relevant behavior encoded in the teacher policy.

C Synthetic recovery of a safety-aware teacher

The main experiments study reward-only MuJoCo locomotion policies. To test whether synthetic recovery is limited to this setting, we include a breadth check on a constrained-control domain from Safety-Gymnasium. Safety-Gymnasium provides safety-critical reinforcement-learning environments with both vector and vision inputs, together with OmniSafe, a library of safe-RL algorithms. We use SafetyPointGoal1, a compact vector-observation navigation task in which the agent must reach a goal while avoiding hazards. Unlike the MuJoCo locomotion policies in the main text, the teacher is trained with PPO-Lag, a Lagrangian variant of PPO that controls constraint violations through a learned Lagrange multiplier [21].

This experiment is not intended to evaluate synthetic distillation as a safe-RL algorithm. The student is not trained with rewards, costs, constraints, or environment interaction. As in the main experiments, the student receives only independently sampled synthetic observations labeled by the fixed teacher policy. The purpose is narrower: to test whether synthetic recovery persists when the teacher was trained under a reward-cost objective and the observation contains explicit safety-relevant structure.

Setup. We train PPO-Lag teachers on SafetyPointGoal1 across five seeds and distill students from teacher-labeled synthetic observations using the same supervised regression objective as Section 2. Each student is evaluated by executing the learned policy in the original SafetyPointGoal1 environment. We report episodic reward, episodic cost, and empirical Lipschitz estimates as mean $\pm$ standard deviation across the five teacher seeds. Reward recovery is computed relative to the corresponding PPO-Lag teacher Cost is reported directly and as a difference from the teacher: $\Delta C = C(\pi_S) - C(\pi_T)$. Negative values indicate lower episodic cost than the teacher. For reward recovery and cost difference, we compute the paired quantity within each seed before averaging across seeds.

Aggregate recovery. Table 6 reports aggregate performance across all synthetic distributions, sample budgets, widths, and seeds. For the student row, each seed is first averaged over all synthetic configurations, and the reported standard deviation is then computed across the five seeds. Synthetic students recover nearly all teacher reward while maintaining lower episodic cost on average. This shows that the recovery phenomenon is not restricted to unconstrained MuJoCo locomotion: it also appears for a safety-aware teacher trained with a constrained reward-cost objective.

Table 6: Synthetic recovery of PPO-Lag teachers on SafetyPointGoal1. Values are mean $\pm$ std across five teacher seeds. Reward recovery and Δ cost are computed relative to the corresponding PPO-Lag teacher within each seed. Cost is episodic cost; lower is better.

Policy group	Recovery (%)	Cost	Δ cost
PPO-Lag teacher	100.0	55.70 $\pm$ 9.60	0.00
Synthetic students	98.68 $\pm$ 1.91	50.33 $\pm$ 2.38	-5.37 $\pm$ 8.45

Across five seeds, PPO-Lag teachers obtain reward 21.72 ± 2.73 and empirical Lipschitz estimate 25.07 ± 2.07 . Averaged over all synthetic student configurations, students obtain reward 21.45 ± 2.89 and empirical Lipschitz estimate 37.70 ± 2.48 . Thus, the average synthetic student recovers 98.68% of teacher reward while reducing mean episodic cost by 5.37 relative to the teacher. This is a strong recovery result because the student has no access to costs, constraints, rewards, transitions, or valid trajectories during training. The only supervision is the teacher’s action on independently sampled synthetic observations.

Representative configurations. Table 7 shows representative high-performing configurations averaged across five teacher seeds. These rows illustrate that reward recovery and cost preservation can occur simultaneously; they are not used as the primary aggregate claim.

Table 7: Representative SafetyPointGoal1 synthetic-student configurations.

Configuration	Recovery (%)	Cost	Δ cost
Gauss., 10k, w=0.25	107.84 ± 4.92	47.36 ± 6.66	-8.34 ± 9.74
Gauss. wide, 50k, w=0.5	106.28 ± 7.00	46.05 ± 3.19	-9.65 ± 7.48
Laplace, 100k, w=0.5	102.28 ± 5.05	43.97 ± 7.64	-11.73 ± 2.67
Uniform wide, 25k, w=0.5	99.15 ± 5.23	37.59 ± 6.17	-18.11 ± 13.74

The corresponding raw rewards for these four configurations are 23.35 ± 2.50 , 22.93 ± 1.63 , 22.15 ± 2.40 , and 21.63 ± 3.57 , respectively. Their empirical Lipschitz estimates are 13.02 ± 1.03 , 32.76 ± 2.82 , 37.54 ± 1.92 , and 46.40 ± 3.31 .

These results suggest that the learned PPO-Lag policy contains recoverable structure in its observation-to-action map, just as the MuJoCo teachers do. The result is especially relevant because the teacher’s behavior depends on safety-relevant observation channels, yet synthetic supervision recovers both task performance and cost behavior.

At the same time, this experiment should not be interpreted as a safe-RL algorithm. The student does not optimize a constraint objective, does not observe costs, and does not receive safety feedback. Instead, it inherits safety-relevant behavior only insofar as that behavior is encoded in the teacher’s action function and recoverable from synthetic queries. Thus, the experiment provides breadth evidence for synthetic policy recovery, not a claim that synthetic distillation enforces safety constraints by itself.

D Experimental Details

This appendix documents hyperparameters, seed counts, compute resources, and full numerical results underlying the figures and tables in the main paper. It also contains three supporting analyses: per-distribution disaggregation of smoothness metrics (D.6), per-environment extensions of the Humanoid-only Tables 1 and 2 from the main text (D.8, D.4), a comparison of student sensitivity on synthetic vs. real inputs (D.11), and a concentration-of-measure analysis explaining why uniform-wide synthetic distributions succeed in high-dimensional environments (D.13). All values are extracted from `.hydra/config.yaml` logs, source code, and existing checkpoints; no new training was performed for any appendix computation.

D.1 Hyperparameters

All four environments use Stable-Baselines3 PPO with the algorithm defaults shown below. The teacher is a shared-trunk MLP with separate actor and critic heads. Both heads use orthogonal initialization with $\text{std} = \sqrt{2}$ for hidden layers, $\text{std} = 0.01$ for the actor output, and $\text{std} = 1.0$ for the critic output.

Observation and reward normalization are enabled through the stored `VecNormalize` statistics used during teacher training and inference. SB3 PPO uses `vf_coef = 0.5` and `max_grad_norm = 0.5`; neither is overridden.

Students are trained by supervised regression of the teacher’s deterministic action mean. The loss is mean-squared error on $(\pi_S(o), \pi_T(o))$ over the synthetic dataset, with no auxiliary terms.

For all main-text mixing experiments and for all appendix tables in this document, only the full-width ($1.0\times$) student is reported. The reduced-width variants are part of a separate ablation and do not affect any conclusion in the main paper. All experiments were run on a single Apple M3 Max (CPU/MPS, no discrete GPU) using PyTorch 2.4.1.

Synthetic observations are sampled in normalized observation space. The teacher’s stored running statistics $(\mu_{\text{rms}}, \sigma_{\text{rms}})$ from `VecNormalize` are used to map between raw and normalized observations: synthetic samples are generated as normalized vectors and consumed directly by the policy network; during evaluation rollouts in the nominal environment, raw observations are mapped to normalized space via $x = (o - \mu_{\text{rms}})/\sigma_{\text{rms}}$ before policy inference.

Table 8: PPO hyperparameters per environment. Entries not listed are SB3 defaults.

Hyperparameter	Hopper-v5	Walker2d-v5	Ant-v5	Humanoid-v5
Clip range	0.2	0.2	0.2	0.2
Learning rate	3×10^{-4}	3×10^{-4}	3×10^{-4}	3×10^{-4}
n_{steps} (rollout)	2,048	2,048	2,048	1,024
PPO epochs	10	10	10	10
Batch size	256	256	4,096	2,048
Entropy coefficient	0.0	0.0	0.0	0.0
Value coefficient	0.5	0.5	0.5	0.5
GAE λ	0.95	0.95	0.95	0.95
γ (discount)	0.99	0.99	0.99	0.99
Hidden sizes	[64, 64]	[64, 64]	[256, 256]	[512, 512]
Activation	Tanh	Tanh	Tanh	Tanh
Total timesteps	2×10^6	2×10^6	1×10^7	1×10^7
Parallel envs	4	4	16	32
Obs. normalization	on	on	on	on
Reward scaling	on	on	on	on
Eval freq. (steps)	10^5	10^5	10^5	10^5

Table 9: Student distillation hyperparameters.

Hyperparameter	Value
Optimizer	Adam (default $\beta_1 = 0.9$, $\beta_2 = 0.999$)
Learning rate	3×10^{-4}
Batch size	2,048
Training epochs	100
Loss function	MSE on deterministic action mean
Gradient clipping	$\ \nabla\ _2 \leq 0.5$
Architecture	2-layer MLP; width $\in \{1.0\times, 0.5\times, 0.25\times\}$ teacher width
Initialization	Orthogonal (std = $\sqrt{2}$ hidden, 0.01 output)
Activation	Tanh
Obs. normalization	on
Sample budgets	{10,000, 25,000, 50,000, 100,000, 250,000, 500,000, 1,000,000, 2,000,000}

D.2 Full numerical results for Figure 1

This subsection reports the underlying recovery percentages for the synthetic distillation curves in Figure 1. Values are normalized to the teacher’s mean return (not to any other condition); the teacher’s per-environment return is shown in each subtable header. Entries are mean $\pm$ std across 10 student seeds trained on full-width architectures. The tables below report full-width recovery curves. Hopper-v5 saturates near 95% at the 100k budget. Walker2d-v5 is shown through 500k samples, where Gaussian, Laplace, and uniform students reach teacher-level recovery. Figure 1 in the main text additionally shows recovery up to 2M samples for Ant-v5 and Humanoid-v5 from upscaled checkpoints.

Table 10: Hopper-v5 recovery (% , mean $\pm$ std) across 10 seeds. Teacher mean return = 1706 ± 617 .

Distribution	10k	25k	50k	100k
Gaussian, $\sigma = 1$	38.5 ± 9.4	81.7 ± 14.5	84.1 ± 14.8	93.7 ± 8.5
Gaussian, $\sigma = 2$ (wide)	44.6 ± 10.8	77.2 ± 23.6	81.3 ± 17.5	88.8 ± 15.5
Gaussian, $\sigma = 0.5$ (narrow)	33.8 ± 11.0	59.7 ± 15.9	73.2 ± 16.0	79.2 ± 17.4
Gaussian, $\mu = 1$, $\sigma = 1$ (shifted)	22.8 ± 4.3	46.2 ± 11.1	63.7 ± 13.7	89.4 ± 26.9
Uniform $[-1, 1]$ (wide)	33.2 ± 8.7	65.1 ± 22.5	78.7 ± 7.6	85.8 ± 16.8
Uniform $[-0.5, 0.5]$ (narrow)	29.4 ± 12.6	34.4 ± 8.4	46.9 ± 16.0	66.6 ± 20.1
Laplace, $b = 1$	45.8 ± 11.5	79.4 ± 20.2	86.4 ± 10.5	96.9 ± 15.7

These are pure synthetic distillation results without real-data mixing; mixing results are reported in Appendix D.12.

Table 11: Walker2d-v5 recovery (% , mean $\pm$ std) across 10 seeds. Teacher mean return = 3181 $\pm$ 1216.

Distribution	10k	25k	50k	100k	250k	500k
$\mathcal{N}(0, 1)$	13.2 $\pm$ 5.2	39.0 $\pm$ 16.5	61.6 $\pm$ 28.9	83.5 $\pm$ 33.6	107.4 $\pm$ 38.4	108.6 $\pm$ 40.0
$\mathcal{N}(0, 2)$	13.8 $\pm$ 5.5	29.5 $\pm$ 14.4	49.7 $\pm$ 26.3	69.5 $\pm$ 34.5	109.1 $\pm$ 38.9	110.6 $\pm$ 39.6
$\mathcal{N}(0, 0.5)$	12.6 $\pm$ 4.9	17.1 $\pm$ 7.3	46.8 $\pm$ 25.0	65.7 $\pm$ 43.9	100.6 $\pm$ 39.9	99.9 $\pm$ 38.5
$\mathcal{N}(1, 1)$	8.3 $\pm$ 8.3	11.3 $\pm$ 3.5	18.2 $\pm$ 7.6	30.8 $\pm$ 13.5	68.8 $\pm$ 47.0	89.5 $\pm$ 45.1
$\mathcal{U}[-1, 1]$	13.2 $\pm$ 5.2	29.4 $\pm$ 17.2	46.3 $\pm$ 26.3	60.6 $\pm$ 40.2	101.8 $\pm$ 41.9	108.9 $\pm$ 39.7
$\mathcal{U}[-0.5, 0.5]$	11.2 $\pm$ 4.3	13.3 $\pm$ 5.5	14.9 $\pm$ 5.6	27.4 $\pm$ 12.7	89.4 $\pm$ 51.4	96.8 $\pm$ 33.9
$\mathcal{L}(0, 1)$	13.4 $\pm$ 4.7	31.7 $\pm$ 15.7	51.6 $\pm$ 17.6	83.9 $\pm$ 37.8	106.3 $\pm$ 41.0	108.1 $\pm$ 41.1

Table 12: Ant-v5 recovery (% , mean $\pm$ std) across 10 seeds. Teacher mean return = 4938 $\pm$ 957.

Distribution	10k	25k	50k	100k
$\mathcal{N}(0, 1)$	46.4 $\pm$ 20.3	57.0 $\pm$ 20.2	72.8 $\pm$ 15.4	81.9 $\pm$ 9.6
$\mathcal{N}(0, 2)$	51.9 $\pm$ 14.2	55.4 $\pm$ 19.2	64.5 $\pm$ 19.2	78.8 $\pm$ 13.3
$\mathcal{N}(0, 0.5)$	36.2 $\pm$ 21.3	45.3 $\pm$ 21.7	61.0 $\pm$ 18.8	77.6 $\pm$ 14.9
$\mathcal{N}(1, 1)$	33.6 $\pm$ 17.6	34.5 $\pm$ 18.0	37.2 $\pm$ 22.7	39.2 $\pm$ 22.4
$\mathcal{U}[-1, 1]$	38.0 $\pm$ 21.9	47.8 $\pm$ 23.4	71.9 $\pm$ 13.1	79.5 $\pm$ 6.8
$\mathcal{U}[-0.5, 0.5]$	21.5 $\pm$ 15.4	25.2 $\pm$ 20.8	42.6 $\pm$ 18.5	62.9 $\pm$ 16.7
$\mathcal{L}(0, 1)$	49.7 $\pm$ 16.7	59.4 $\pm$ 17.6	70.0 $\pm$ 13.8	80.8 $\pm$ 11.0

Table 13: Humanoid-v5 recovery (% , mean $\pm$ std) across 10 seeds. Teacher mean return = 6673.42 $\pm$ 757.35.

Distribution	10k	25k	50k	100k	250k	500k	1M	2M
$\mathcal{N}(0, 1)$	34.5 $\pm$ 8.6	35.0 $\pm$ 10.2	35.5 $\pm$ 11.9	36.4 $\pm$ 11.5	44.3 $\pm$ 17.2	67.9 $\pm$ 23.0	69.8 $\pm$ 24.3	77.9 $\pm$ 18.9
$\mathcal{N}(0, 2)$	30.3 $\pm$ 6.5	27.1 $\pm$ 5.4	27.4 $\pm$ 8.2	31.6 $\pm$ 8.2	36.6 $\pm$ 15.3	47.4 $\pm$ 25.2	67.4 $\pm$ 26.6	69.5 $\pm$ 29.2
$\mathcal{N}(0, 0.5)$	37.0 $\pm$ 13.0	38.1 $\pm$ 13.3	40.8 $\pm$ 13.3	45.0 $\pm$ 15.2	49.1 $\pm$ 23.6	64.7 $\pm$ 27.5	71.6 $\pm$ 26.6	72.0 $\pm$ 23.3
$\mathcal{N}(1, 1)$	22.5 $\pm$ 13.4	23.2 $\pm$ 14.7	24.8 $\pm$ 13.4	27.8 $\pm$ 15.4	33.9 $\pm$ 19.6	48.1 $\pm$ 25.4	38.2 $\pm$ 22.4	47.2 $\pm$ 26.2
$\mathcal{U}[-1, 1]$	34.7 $\pm$ 10.1	38.8 $\pm$ 12.3	39.1 $\pm$ 11.2	44.9 $\pm$ 15.2	47.5 $\pm$ 22.6	64.4 $\pm$ 25.1	69.1 $\pm$ 24.9	74.3 $\pm$ 20.9
$\mathcal{U}[-0.5, 0.5]$	40.3 $\pm$ 14.8	40.4 $\pm$ 17.7	42.6 $\pm$ 15.3	43.2 $\pm$ 15.1	47.6 $\pm$ 21.1	51.3 $\pm$ 24.6	64.3 $\pm$ 25.9	63.5 $\pm$ 25.8
$\mathcal{L}(0, 1)$	35.8 $\pm$ 11.4	30.3 $\pm$ 7.6	30.0 $\pm$ 7.6	33.9 $\pm$ 7.1	40.3 $\pm$ 18.7	62.1 $\pm$ 28.2	72.1 $\pm$ 26.7	77.3 $\pm$ 22.0

The coordinate-wise shifted Gaussian is a deliberately off-centered stress test, but it also increases aggregate input norm in high-dimensional spaces. Table 14 controls for this by shifting the mean along a random direction while keeping the expected squared norm equal to that of $x \sim \mathcal{N}(0, I_d)$. The milder $c = 0.50$ shift often matches the centered Gaussian baseline, indicating that small off-center shifts are not necessarily harmful. In contrast, the larger $c = 0.75$ shift consistently underperforms the centered Gaussian baseline across Walker2d-v5, Ant-v5, and Humanoid-v5. Thus, shifted-distribution failure is not solely attributable to larger aggregate input norm or activation saturation; distributional placement still matters.

D.3 Sensitivity Structure Under Synthetic Probing

This section reports the full sensitivity-structure analysis underlying Table 1 of the main text. We compare the input-sensitivity geometry of the teacher policy as measured on real rollout observations versus synthetic inputs drawn from various distributions in normalized observation space.

Setup. For each teacher policy we compute $G_\pi = \frac{1}{n} \sum_i J_i^\top J_i$, where $J_i = \partial \pi(x_i) / \partial x_i$ is evaluated at inputs x_i . For real observations, x_i are sampled from rollout trajectories. For synthetic probing, $x_i \sim q(x)$ where q is a synthetic distribution in normalized observation space. All Jacobians use autograd on the normalized actor network (teacher_sb3.zip); 1000 probe points per seed for the main tables, 10000 for the robustness check below.

Robustness to probe count. Table 15 verifies that the sensitivity estimates are stable across probe counts. Increasing from 1000 to 10000 real observations per seed shifts r_{90} and r_{95} slightly but leaves d_{sens} and the qualitative interpretation unchanged.

Table 14: Norm-controlled random-shift control. Values are recovery percentages, mean $\pm$ std across 10 seeds. All distributions are defined in normalized policy-input space. The norm-controlled distribution samples $x \sim \mathcal{N}(\alpha u, \sigma^2 I_d)$ with $\alpha = c\sqrt{d}$ and $\sigma = \sqrt{1 - c^2}$, where u is a random unit direction, so that $\mathbb{E}\|x\|_2^2 = d$, matching the standard Gaussian baseline.

Environment	Budget	Gaussian	Original shifted	Norm-shift $c = 0.50$	Norm-shift $c = 0.75$
Walker2d-v5	100k	88.3 ± 17.1	37.7 ± 20.3	86.7 ± 26.9	36.2 ± 22.5
Walker2d-v5	500k	98.2 ± 6.6	85.5 ± 40.8	99.7 ± 11.0	81.9 ± 27.5
Ant-v5	100k	81.9 ± 8.3	40.0 ± 24.0	66.8 ± 19.4	34.2 ± 17.9
Ant-v5	2M	94.0 ± 3.9	69.3 ± 14.1	98.0 ± 7.7	72.2 ± 18.0
Humanoid-v5	100k	35.8 ± 8.6	27.1 ± 13.1	38.5 ± 11.7	28.2 ± 17.5
Humanoid-v5	2M	80.9 ± 12.8	48.7 ± 23.9	77.7 ± 17.5	70.9 ± 22.2

Table 15: Humanoid-v5 teacher sensitivity at 1000 vs. 10000 real rollout observations per seed. Mean $\pm$ std across 10 seeds.

Probe set	r_{90}	r_{95}	r_{99}	d_{sens}
1000 obs/seed	77.9 ± 21.1	168.4 ± 21.5	296.9 ± 8.7	15.58 ± 1.05
10000 obs/seed	82.1 ± 22.8	174.8 ± 22.7	300.5 ± 8.6	15.68 ± 1.16

Per-environment teacher sensitivity spectrum. Table 16 reports the teacher-side sensitivity spectrum across all four environments. The $d_{\text{sens}}/d_{\text{obs}}$ ratio decreases sharply with ambient dimension, confirming that higher-dimensional environments are more anisotropic in relative terms.

Table 16: Per-environment teacher sensitivity spectrum. r_{90} and r_{95} are the numbers of eigendirections capturing 90% and 95% of total Jacobian variance; d_{sens} is the participation-ratio sensitivity dimension. Mean $\pm$ std across 10 seeds. All values use normalized observations and autograd Jacobians.

Environment	d_{obs}	r_{90}	r_{95}	d_{sens}	$d_{\text{sens}}/d_{\text{obs}}$
Hopper-v5	11	4.2 ± 0.6	5.5 ± 0.7	3.46 ± 0.53	0.315
Walker2d-v5	17	9.0 ± 0.7	11.7 ± 0.7	5.91 ± 0.80	0.348
Ant-v5	105	15.4 ± 2.0	30.4 ± 3.7	8.29 ± 0.73	0.079
Humanoid-v5	348	77.8 ± 21.6	168.3 ± 22.4	15.50 ± 1.10	0.045

Cross-environment real vs. synthetic alignment. Table 17 extends the two-column alignment analysis of main-text Table 1 across all four environments. The $k = r_{90}$ alignment floor is an environment-level property: it varies from ≈ 0.98 on Hopper to ≈ 0.76 on Humanoid, and within each environment no centered distribution family escapes it.

Humanoid student sensitivity structure. Table 18 reports how the student’s sensitivity spectrum evolves with sample count. Additional synthetic samples recover progressively more of the teacher’s sensitivity tail, but r_{90} for the student (~ 29 at 2M samples) remains substantially below the teacher (~ 78), consistent with the $\cos \bar{\theta} \approx 0.76$ tail-alignment floor in main-text Table 1.

D.4 Per-environment subspace overlap

Top- k overlap at $k = \lceil d_{\text{sens}} \rceil$. Table 19 reports teacher–student subspace overlap at the participation-ratio dimension across environments. Overlap is high in all cases. For Hopper, where d_{sens} nearly equals the ambient dimension, this is essentially trivial. For Walker2d, Ant, and Humanoid, high overlap provides evidence that students recover the teacher’s dominant sensitivity geometry rather than a low-dimensional approximation aligned by chance.

Broader subspace at $k = r_{90}$. Table 20 reports alignment at the r_{90} subspace for Humanoid-v5, together with reference baselines needed to interpret the $\cos \bar{\theta} \approx 0.76$ value reported in main-text Table 1.

Table 17: Sensitivity structure under real and synthetic inputs across environments. $\cos \bar{\theta}$ measures alignment with the real-observation teacher eigenspace at two scales: $k = \lceil d_{\text{sens}}(\text{real}) \rceil$ and $k = r_{90}(\text{real})$. Mean $\pm$ std across 10 seeds.

Env.	Source	d_{sens}	$\cos \bar{\theta}$ (top- k)	$\cos \bar{\theta}$ (r_{90})
Hopper	Real	3.46 ± 0.53	1.000	1.000
	Gaussian	3.62 ± 0.43	0.986 ± 0.010	0.984 ± 0.021
	Uniform	3.23 ± 0.32	0.987 ± 0.009	0.984 ± 0.017
	Shifted	3.62 ± 0.70	0.967 ± 0.027	0.960 ± 0.035
Walker2d	Real	5.91 ± 0.80	1.000	1.000
	Gaussian	4.90 ± 0.53	0.924 ± 0.035	0.951 ± 0.017
	Uniform	4.39 ± 0.47	0.914 ± 0.043	0.947 ± 0.020
	Shifted	5.02 ± 1.03	0.918 ± 0.030	0.943 ± 0.025
Ant	Real	8.29 ± 0.73	1.000	1.000
	Gaussian	10.41 ± 1.10	0.963 ± 0.009	0.950 ± 0.007
	Uniform	7.43 ± 0.68	0.957 ± 0.015	0.952 ± 0.007
	Shifted	12.50 ± 1.72	0.917 ± 0.016	0.911 ± 0.008
Humanoid	Real	15.60 ± 1.22	1.000	1.000
	Gaussian	17.52 ± 0.80	0.988 ± 0.003	0.764 ± 0.034
	Uniform	12.71 ± 0.55	0.988 ± 0.003	0.766 ± 0.032
	Shifted	21.48 ± 1.11	0.950 ± 0.007	0.724 ± 0.033

Table 18: Humanoid-v5 input-sensitivity structure of synthetic students across 10 seeds. $G_{\pi} = n^{-1} \sum_i J_i^{\top} J_i$ computed on 1000 real observations per seed.

Distribution	Samples	r_{90}	r_{95}	d_{sens}
Teacher	–	77.8 ± 21.6	168.3 ± 22.4	15.50 ± 1.10
Gaussian baseline	25k	16.2 ± 0.6	55.5 ± 16.2	12.19 ± 0.46
Gaussian narrow	25k	14.8 ± 0.6	22.1 ± 7.3	11.52 ± 0.41
Gaussian shifted	25k	15.9 ± 0.7	56.4 ± 17.2	12.02 ± 0.48
Gaussian wide	25k	23.4 ± 7.4	93.1 ± 16.9	13.09 ± 0.62
Gaussian baseline	500k	16.7 ± 1.1	51.2 ± 13.6	12.37 ± 0.72
Gaussian narrow	500k	16.1 ± 0.5	41.2 ± 11.8	12.53 ± 0.59
Uniform wide	500k	16.2 ± 0.6	46.8 ± 12.0	12.64 ± 0.64
Laplace baseline	500k	15.9 ± 0.8	40.8 ± 10.4	11.68 ± 0.69
Gaussian baseline	2M	29.0 ± 10.1	96.6 ± 20.0	13.36 ± 0.84
Gaussian narrow	2M	26.8 ± 10.1	93.0 ± 19.4	13.75 ± 0.83
Uniform wide	2M	29.8 ± 10.8	97.4 ± 20.3	13.86 ± 0.85
Laplace baseline	2M	28.4 ± 10.4	95.3 ± 21.1	13.07 ± 0.85

The random-subspace baseline (0.411) gives the expected alignment of a randomly oriented k -dimensional subspace of the same dimension in a 348-dimensional space. The teacher self-consistency ceiling (0.929) gives the alignment between two independent estimates of the teacher’s r_{90} eigenspace from split-half rollout observations, representing the highest alignment achievable given finite-sample estimation noise. The student alignment (0.598 unweighted, 0.520 variance-weighted) sits 36% of the way from random to ceiling at 10^5 samples, indicating partial but incomplete recovery of the sensitivity tail.

The variance-weighted overlap (0.520) weights each eigendirection by its teacher eigenvalue λ_T , giving more weight to directions that carry more sensitivity. That it falls modestly below the unweighted measure (0.598) indicates the alignment gap is distributed across the subspace rather than concentrated in low-eigenvalue tail directions. The eigenvalue ratio $\lambda_{16}/\lambda_{78} \approx 22$ confirms that top-16 directions carry substantially more sensitivity than ranks 17–78, but the r_{90} subspace is itself stably identifiable from finite samples (self-consistency 0.929), ruling out estimation noise as the explanation for the gap.

Table 19: Teacher–student subspace overlap at $k = \lceil d_{\text{sens}} \rceil$ across environments. $\cos \bar{\theta}$ is the mean cosine of principal angles. 10 seeds per condition; Gaussian baseline at 10^5 samples.

Environment	$k = \lceil d_{\text{sens}} \rceil$	$\cos \bar{\theta}$
Hopper-v5	4	0.999 ± 0.001
Walker2d-v5	6	0.950 ± 0.030
Ant-v5	7	0.982 ± 0.008
Humanoid-v5	16	0.986 ± 0.003

Table 20: Humanoid-v5 subspace overlap at $k = r_{90} = 78$. Teacher self-consistency is computed via split-half estimation on rollout observations from each seed (10 seeds). Random baseline is averaged over 10 random orthonormal bases versus the teacher subspace.

Quantity	$\cos \bar{\theta}$	Interpretation
Random-subspace baseline	0.411 ± 0.001	noise floor
Student vs. teacher (unweighted)	0.598 ± 0.005	observed
Student vs. teacher (variance-weighted)	0.520 ± 0.005	weighted by λ_T
Teacher self-consistency	0.929 ± 0.018	ceiling

D.5 Subspace intervention: full numerical results

Sampling procedure. All sampling occurs in normalized observation space $x = (o - \mu_{\text{rms}})/\sigma_{\text{rms}}$. For each condition we specify an orthonormal basis $V_k \in \mathbb{R}^{d \times k}$ and sample $z \sim \mathcal{N}(0, I_k)$, setting $x = V_k z$. We do not rescale by d/k : per-active-direction variance is held at unity across all conditions so that only the support directions differ. The teacher’s empirical sensitivity matrix $G_\pi = n^{-1} \sum_i J_i^\top J_i$ is computed in normalized space using $n = 1000$ real teacher-rollout observations per seed; eigendecomposition gives the top- k , bottom- k , and full eigenvector bases. Random subspaces are sampled by QR-decomposing a random $d \times k$ Gaussian matrix.

Conditions. Humanoid-v5 ($d_{\text{sens}} \approx 16$, $r_{90} \approx 78$): Full $\mathcal{N}(0, I_{348})$, Top-16, Random-16, Bottom-16, Top-78, Random-78. Ant-v5 ($d_{\text{sens}} \approx 7$, $r_{90} \approx 17$): Full $\mathcal{N}(0, I_{105})$, Top-7, Random-7, Bottom-7, Top-17, Random-17. We do not test Bottom-78 or Bottom-17 because in higher- k regimes the bottom subspace contains more variance than the lower- k top subspace and the comparison becomes uninformative.

Training and evaluation. Main-text results use 20 training epochs. We use 20 epochs for the subspace conditions because longer training induces overfitting to the restricted input geometry: students trained strictly within a subspace have no supervision off it, and additional epochs tighten the fit inside the subspace without improving — and often worsening — closed-loop behavior in the full observation space. Table 21 reports results at both 20 and 100 epochs; the directional ordering Top- $k > \text{Random-}k > \text{Bottom-}k$ is preserved at both epoch counts, confirming that the causal claim does not depend on the epoch choice. Evaluation uses 20 rollouts per student in the nominal environment.

Table 21: Subspace intervention recovery at 20 vs. 100 training epochs. Recovery (%), mean $\pm$ std across 10 seeds. All conditions use 10^5 samples. 100-epoch runs are independent stochastic training runs, not continuations of the 20-epoch students.

Condition	Humanoid-v5		Ant-v5	
	20 ep.	100 ep.	20 ep.	100 ep.
Full	39.8 ± 18.0	32.9 ± 16.8	81.0 ± 11.2	91.3 ± 11.6
Top- d_{sens}	16.4 ± 9.9	14.4 ± 5.0	25.8 ± 13.1	26.3 ± 9.3
Random- d_{sens}	3.6 ± 1.8	3.3 ± 1.5	18.7 ± 2.5	17.2 ± 3.4
Bottom- d_{sens}	2.8 ± 0.8	3.2 ± 1.4	16.9 ± 4.3	18.0 ± 2.3
Top- r_{90}	25.0 ± 9.1	14.9 ± 5.8	39.6 ± 19.3	28.2 ± 15.8
Random- r_{90}	4.5 ± 1.3	3.9 ± 1.2	16.5 ± 3.7	9.9 ± 3.5

Sensitivity-overlap vs. recovery. Table 22 reports the normalized sensitivity-overlap score $\text{Tr}(G_\pi \Sigma_q) / \text{Tr}(G_\pi)$ for each condition, computed analytically from the teacher spectrum, alongside recovery at 20 epochs. The overlap score ranks all six Humanoid conditions monotonically (Spearman $\rho = 1.00$ across condition means) and strongly on Ant ($\rho = 0.83$).

Table 22: Normalized sensitivity-overlap score and recovery for each subspace condition. Overlap computed analytically; recovery is mean $\pm$ std across 10 seeds at 20 training epochs.

Condition	Humanoid-v5		Ant-v5	
	Overlap	Recovery (%)	Overlap	Recovery (%)
Bottom- d_{sens}	0.003 ± 0.000	2.8 ± 0.9	0.002 ± 0.001	16.9 ± 4.5
Random- d_{sens}	0.046 ± 0.000	3.6 ± 1.9	0.067 ± 0.000	18.7 ± 2.6
Random- r_{90}	0.224 ± 0.000	4.5 ± 1.4	0.162 ± 0.000	16.5 ± 3.9
Top- d_{sens}	0.810 ± 0.023	16.4 ± 10.4	0.736 ± 0.042	25.8 ± 13.8
Top- r_{90}	0.901 ± 0.014	25.0 ± 9.6	0.897 ± 0.031	39.6 ± 20.3
Full	1.000 ± 0.000	39.8 ± 19.0	1.000 ± 0.000	81.0 ± 11.8
Spearman ρ (condition means): Humanoid 1.00, Ant 0.83				

Walker2d-v5: regime boundary. The subspace intervention is most discriminating when the sensitive subspace constitutes a small fraction of the ambient input space, so that top- k and random- k conditions differ substantially in sensitivity overlap while still providing enough dimensionality for the student to learn a viable policy. Walker2d-v5 ($d_{\text{obs}} = 17$, $d_{\text{sens}} \approx 6$, $r_{90} \approx 9$) does not satisfy this condition: projecting to any subspace of dimension ≤ 9 removes a large fraction of the input space, collapsing recovery regardless of direction. Table 23 reports the Walker2d results for completeness; Top-6 and Random-6 are statistically indistinguishable given the standard deviations, making the comparison uninformative about the directional hypothesis in this regime.

Table 23: Walker2d-v5 subspace intervention. Recovery (%), mean $\pm$ std across 10 seeds, 100 training epochs. Top-6 and Random-6 are not statistically distinguishable.

Condition	Overlap / $\text{Tr}(G)$	Recovery (%)	Return
Full	1.000 ± 0.000	99.5 ± 14.9	3463 ± 1273
Top-9	0.910 ± 0.014	21.3 ± 17.2	712 ± 715
Random-9	0.529 ± 0.000	13.3 ± 20.8	369 ± 418
Top-6	0.825 ± 0.023	4.1 ± 4.4	169 ± 162
Random-6	0.353 ± 0.000	3.5 ± 5.6	118 ± 173
Bottom-6	0.055 ± 0.011	3.2 ± 3.8	108 ± 118

D.6 Per-distribution disaggregation of smoothness metrics

The main-text Table 5 reports teacher and student smoothness metrics aggregated across synthetic sampling distributions. Aggregation can mask the relationship between distribution choice and student smoothness; in particular, the shifted-distribution students discussed in Section 4.4 of the main text are smoother than baseline-distribution students yet recover less performance. The following tables disaggregate the analysis by distribution. All values are mean $\pm$ SEM across 10 seeds for full-width 100k-sample students. Lipschitz constants are mean operator norms of the policy Jacobian on real rollout observations; gradient norms are reported in units of 10^{-5} .

The shifted distribution (Gaussian with $\mu = 1$) consistently produces students with the lowest Lipschitz constants and gradient norms in every environment, yet also the lowest recovery percentages. This confirms the main-text observation that smoothness alone is insufficient: a student can be smooth and still fail if its smoothness is not aligned with the teacher’s sensitive directions.

D.7 Per-environment sensitivity spectrum

The main-text Table 1 reports the participation-ratio sensitivity dimension d_{sens} for Humanoid-v5. Table 27 reports the same teacher-side quantity across environments, computed using the eigenvalue-based participation ratio of $G_\pi = (1/n) \sum_i J_i^\top J_i$ on real teacher-rollout observations. Walker2d-v5

Table 24: Hopper-v5 per-distribution smoothness.

Distribution	Recovery (%)	Lipschitz	$\ \text{logit}\ $	Grad. norm ($\times 10^{-5}$)
Teacher	100.0	0.696	1.516	17.07
Gaussian, $\sigma = 1$	65.3 ± 6.9	0.648 ± 0.018	1.476 ± 0.034	15.98 ± 0.38
Gaussian, $\sigma = 2$	60.9 ± 5.5	0.605 ± 0.016	1.393 ± 0.031	15.07 ± 0.35
Gaussian, $\sigma = 0.5$	53.5 ± 6.2	0.646 ± 0.019	1.537 ± 0.045	15.87 ± 0.39
Gaussian, $\mu = 1, \sigma = 1$	42.1 ± 6.2	0.598 ± 0.024	1.325 ± 0.046	14.57 ± 0.60
Uniform $[-1, 1]$	55.2 ± 6.4	0.650 ± 0.019	1.526 ± 0.038	16.03 ± 0.39
Uniform $[-0.5, 0.5]$	34.0 ± 4.1	0.618 ± 0.018	1.589 ± 0.052	15.38 ± 0.32
Laplace, $b = 1$	64.0 ± 6.7	0.637 ± 0.017	1.459 ± 0.032	15.74 ± 0.39

Table 25: Ant-v5 per-distribution smoothness.

Distribution	Recovery (%)	Lipschitz	$\ \text{logit}\ $	Grad. norm ($\times 10^{-5}$)
Teacher	100.0	0.187	0.941	2.49
Gaussian, $\sigma = 1$	56.6 ± 3.9	0.159 ± 0.005	0.889 ± 0.025	2.03 ± 0.04
Gaussian, $\sigma = 2$	53.6 ± 3.3	0.156 ± 0.007	0.865 ± 0.034	1.97 ± 0.07
Gaussian, $\sigma = 0.5$	48.4 ± 3.8	0.172 ± 0.003	0.955 ± 0.023	2.20 ± 0.04
Gaussian, $\mu = 1, \sigma = 1$	33.8 ± 0.9	0.148 ± 0.002	0.811 ± 0.004	1.91 ± 0.04
Uniform $[-1, 1]$	49.9 ± 4.2	0.171 ± 0.003	0.947 ± 0.024	2.20 ± 0.04
Uniform $[-0.5, 0.5]$	35.3 ± 3.1	0.183 ± 0.003	1.032 ± 0.027	2.36 ± 0.05
Laplace, $b = 1$	57.8 ± 3.7	0.154 ± 0.006	0.865 ± 0.028	1.96 ± 0.05

Table 26: Humanoid-v5 per-distribution smoothness.

Distribution	Recovery (%)	Lipschitz	$\ \text{logit}\ $	Grad. norm ($\times 10^{-5}$)
Teacher	100.0	0.255	3.001	4.10
Gaussian, $\sigma = 1$	36.6 ± 0.5	0.257 ± 0.004	3.274 ± 0.036	3.87 ± 0.08
Gaussian, $\sigma = 2$	32.3 ± 0.9	0.261 ± 0.008	3.302 ± 0.081	4.06 ± 0.13
Gaussian, $\sigma = 0.5$	39.5 ± 0.8	0.268 ± 0.003	3.393 ± 0.047	3.89 ± 0.07
Gaussian, $\mu = 1, \sigma = 1$	22.5 ± 1.1	0.203 ± 0.003	2.553 ± 0.028	3.20 ± 0.06
Uniform $[-1, 1]$	39.2 ± 0.8	0.265 ± 0.002	3.355 ± 0.032	3.87 ± 0.06
Uniform $[-0.5, 0.5]$	40.0 ± 0.6	0.271 ± 0.006	3.412 ± 0.087	3.86 ± 0.10
Laplace, $b = 1$	34.5 ± 0.9	0.257 ± 0.006	3.267 ± 0.065	3.95 ± 0.11

is included as a fully replicated medium-dimensional locomotion environment, replacing the previous single-seed locomotion case study.

Table 28 reports the corresponding Walker2d-v5 student-side sensitivity recovery. By 500k samples, Gaussian, Laplace, and uniform-wide students match the teacher’s r_{90} , r_{95} , and d_{sens} within seed-level variation.

D.8 Per-environment coordinate alignment

The main-text Table 4 reports teacher–student coordinate-sensitivity alignment for Humanoid-v5. Table 29 extends this measurement across environments using the same methodology: Spearman rank correlation and cosine similarity between per-coordinate sensitivity vectors $I(\pi_T)$ and $I(\pi_S)$, computed on real teacher-rollout observations.

Coordinate alignment is high across environments, confirming that the recovery of sensitive coordinates documented for Humanoid-v5 in the main text is not a high-dimensional artifact. Table 30 shows that Walker2d-v5 alignment tightens substantially at 500k samples, where Gaussian and Laplace students recover the teacher’s coordinate and subspace sensitivity structure almost exactly.

D.9 Per-environment subspace overlap

Section 4.3 reports subspace overlap between teacher and student top- d_{sens} eigenspaces for Humanoid. We extend this analysis across environments and to the broader r_{90} subspace where applicable.

Table 27: Per-environment teacher sensitivity spectrum. r_{90} and r_{95} are the numbers of eigendirections capturing 90% and 95% of total Jacobian variance; d_{sens} is the participation-ratio sensitivity dimension. Mean $\pm$ std across 10 seeds.

Environment	d_{obs}	r_{90}	r_{95}	d_{sens}	$d_{\text{sens}}/d_{\text{obs}}$	Seeds
Hopper-v5	11	4.2 ± 0.6	5.5 ± 0.7	3.46 ± 0.53	0.315	10
Walker2d-v5	17	9.0 ± 0.7	11.7 ± 0.7	5.91 ± 0.80	0.348	10
Ant-v5	105	16.8 ± 4.1	33.5 ± 6.7	6.93 ± 1.24	0.066	10
Humanoid-v5	348	77.8 ± 21.6	168.3 ± 22.4	15.50 ± 1.10	0.045	10

Table 28: Walker2d-v5 input-sensitivity structure. $G_\pi = n^{-1} \sum_i J_i^\top J_i$ on 1000 real teacher-rollout observations per seed. Mean $\pm$ std across 10 seeds.

Policy	Samples	r_{90}	r_{95}	d_{sens}
Teacher	–	9.00 ± 0.67	11.70 ± 0.67	5.91 ± 0.80
Gaussian baseline	100k	8.00 ± 0.67	10.70 ± 0.67	5.50 ± 1.13
Gaussian baseline	500k	9.10 ± 0.74	11.70 ± 0.67	5.88 ± 0.83
Laplace baseline	500k	9.00 ± 0.67	11.70 ± 0.67	5.94 ± 0.82
Uniform wide	500k	9.00 ± 0.67	11.80 ± 0.63	5.93 ± 0.79

Table 29: Per-environment teacher–student coordinate-sensitivity alignment. Values are mean $\pm$ std or SEM as reported by each analysis.

Environment	Spearman ρ	Cosine sim.
Hopper-v5	0.833 ± 0.024	0.981 ± 0.002
Walker2d-v5	0.84 ± 0.19	0.98 ± 0.03
Ant-v5	0.961 ± 0.003	0.996 ± 0.000
Humanoid-v5	0.830 ± 0.008	0.996 ± 0.000

Subspace overlap at the participation-ratio dimension is high across environments. For Hopper, where d_{sens} nearly equals the ambient dimension, this is essentially trivial. For Walker2d, Ant, and Humanoid, high subspace overlap provides evidence that students recover the teacher’s dominant sensitivity geometry rather than merely a low-dimensional approximation aligned by chance.

Extended subspace at $k = r_{90}$. We additionally test the broader r_{90} subspace, which captures 90% of total Jacobian variance. For Humanoid, $r_{90} = 78$ in a 348-dimensional observation space.

Student-teacher alignment at $k = 78$ sits 36% of the way from random (0.411) to teacher self-consistency (0.929), indicating partial but incomplete recovery of the long-tail sensitivity subspace at 10^5 samples. The variance-weighted overlap (weighting eigendirections by teacher eigenvalues λ_T) gives 0.520, modestly below the unweighted measure—consistent with the gap being spread across the subspace rather than concentrated in low-eigenvalue tail directions. The eigenvalue ratio $\lambda_{16}/\lambda_{78} \approx 22$ confirms that top-16 directions carry substantially more sensitivity than ranks 17–78, but the subspace at $k = 78$ is itself stably identifiable from finite samples (teacher self-consistency $\cos \bar{\theta} = 0.929$), ruling out measurement noise as the explanation for the gap. The budget-dependent recovery pattern reported in Appendix D.7 suggests that students recover increasing amounts of the long-tail sensitivity at larger sample budgets (r_{90} rises from ~ 16 at 10^5 samples to ~ 29 at 2M samples); a corresponding analysis of subspace overlap at higher budgets is left to future work.

D.10 Real-fit Gaussian: decomposition diagnostic

As a diagnostic baseline, we fit a multivariate Gaussian to teacher rollout observations and distill from samples drawn from this distribution. This baseline is real-data-informed and is therefore *not* part of our data-free synthetic recovery claim. We include it to characterize how much of the gap between the data-free recovery percentages of Section 3 and full teacher performance is attributable to distributional structure available only through rollouts.

Table 30: Walker2d-v5 teacher-student sensitivity alignment. Coordinate metrics compare teacher and student coordinate-wise sensitivity vectors, while subspace overlap measures alignment of the top- $\lceil d_{\text{sens}} \rceil$ eigenspaces. Values are mean $\pm$ std across 10 seeds.

Student	Samples	Spearman ρ	Cosine sim.	Subspace $\cos \bar{\theta}$
Gaussian baseline	100k	0.84 ± 0.19	0.98 ± 0.03	0.95 ± 0.03
Gaussian baseline	500k	0.98 ± 0.02	1.00 ± 0.00	0.99 ± 0.02
Laplace baseline	500k	0.99 ± 0.01	1.00 ± 0.00	1.00 ± 0.00
Uniform wide	500k	0.90 ± 0.12	0.99 ± 0.01	0.97 ± 0.01

Table 31: Teacher-student subspace overlap at $k = \lceil d_{\text{sens}} \rceil$ across environments. Mean principal angle in degrees; $\cos \bar{\theta}$ in parentheses where available. 10 seeds per condition; Gaussian baseline at 10^5 samples.

Environment	$k = \lceil d_{\text{sens}} \rceil$	$\cos \bar{\theta}$
Hopper-v5	4	0.999 ± 0.001 (2.1°)
Walker2d-v5	6	0.950 ± 0.030
Ant-v5	7	0.982 ± 0.008 (10.9°)
Humanoid-v5	16	0.986 ± 0.003 (9.7°)

Setup. For each environment we collect $n = 10,000$ real teacher-rollout observations and compute their empirical mean $\hat{\mu}$ and Ledoit-Wolf-shrunk covariance $\hat{\Sigma}$ in normalized observation space. Ledoit-Wolf shrinkage [22] regularizes the covariance estimate toward a scaled identity; for Humanoid ($d = 348$, $n = 10,000$) the analytical shrinkage coefficient is $\alpha \approx 0.03$, indicating the sample covariance is well-conditioned at this sample size. We compare four sampling distributions in a 2×2 factorial design:

- $\mathcal{N}(0, I)$ — the simple Gaussian baseline used throughout the paper (neither corrected center nor empirical covariance);
- $\mathcal{N}(\hat{\mu}, I)$ — centering corrected to empirical rollout mean, isotropic covariance;
- $\mathcal{N}(0, \hat{\Sigma})$ — empirical covariance, zero-centered;
- $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$ — the real-fit Gaussian.

The factorial design isolates the contribution of distributional centering and empirical covariance structure, and tests whether their combination is additive. Tables 33 and 34 report recovery for each of the four conditions on Humanoid and Ant respectively, and Table 35 reports the decomposition into centering, covariance, and interaction effects.

Why centering matters at all. The simple Gaussian $\mathcal{N}(0, I)$ in normalized space is centered at the running-mean normalization statistics $(\mu_{\text{rms}}, \sigma_{\text{rms}})$, which are accumulated as running averages over the entire training history. By design, these statistics lag the converged policy’s current rollout statistics: a `VecNormalize` wrapper that tracked the current rollout mean tightly would induce non-stationary value-function targets and destabilize PPO training. The empirical mean of normalized rollout observations from a converged Humanoid policy has $\|\hat{\mu}\| \approx 4.9$ in 348-dimensional normalized space, reflecting genuine drift between training-history-averaged and converged-policy state distributions. This drift is a feature of the standard normalization design, not a bug. The 2×2 design quantifies how much it costs in recovery.

Interpretation. On both environments, off-diagonal covariance structure dominates the recovery improvement: 66% of the total gap on Humanoid and 89% on Ant. Re-centering on the empirical rollout mean contributes a smaller fraction (13% Humanoid, 24% Ant). The two effects are super-additive on Humanoid (interaction +11.4pp) and approximately independent on Ant (interaction −2.8pp, within noise). The Humanoid super-additivity is consistent with a high-dimensional anisotropic policy: the policy-relevant region is at a particular off-diagonal direction *from* a particular center, and getting both factors right captures more than the sum of their individual contributions.

Table 32: Humanoid-v5 subspace overlap at $k = 78 = r_{90}$, with reference values for interpretation. Teacher self-consistency computed via split-half estimation on rollout observations from each seed (10 seeds). Random baseline averaged over 10 random orthonormal bases versus the teacher subspace.

Quantity	$\cos \bar{\theta}$	Interpretation
Random-subspace baseline	0.411 ± 0.001	noise floor
Student vs. teacher (unweighted)	0.598 ± 0.005	observed
Student vs. teacher (variance-weighted)	0.520 ± 0.005	weighted by λ_T
Teacher self-consistency	0.929 ± 0.018	ceiling

Table 33: Real-fit Gaussian decomposition, Humanoid-v5. Recovery (%) as mean $\pm$ std across 10 seeds.

Sampling distribution	Recovery (%)
$\mathcal{N}(0, I)$	35.9 ± 14.0
$\mathcal{N}(\hat{\mu}, I)$	42.5 ± 16.4
$\mathcal{N}(0, \hat{\Sigma})$	70.6 ± 19.8
$\mathcal{N}(\hat{\mu}, \hat{\Sigma})$	88.7 ± 11.0

Table 34: Real-fit Gaussian decomposition, Ant-v5. Recovery (%) as mean $\pm$ std across 10 seeds.

Sampling distribution	Recovery (%)
$\mathcal{N}(0, I)$	78.8 ± 19.9
$\mathcal{N}(\hat{\mu}, I)$	83.8 ± 11.7
$\mathcal{N}(0, \hat{\Sigma})$	97.6 ± 5.9
$\mathcal{N}(\hat{\mu}, \hat{\Sigma})$	99.9 ± 12.7

The decomposition has direct relevance to the data-free claim. The `obs_rms` normalization captures per-dimension scale but not inter-coordinate correlations; the convention of using $\mathcal{N}(0, I)$ in normalized space therefore samples from a distribution that is correctly scaled per-dimension but does not respect off-diagonal state structure. The real-fit experiment quantifies the cost of this restriction. On Humanoid, the cost is roughly 50 percentage points of recovery; the bulk of this cost (35pp) comes from not capturing the empirical covariance, which is unavailable from policy metadata alone.

D.11 Student sensitivity on synthetic vs. real inputs

Tables 24, 25, 26 and 29 measure student geometry on real teacher-rollout observations. Because students are trained exclusively on synthetic inputs, it is also informative to ask whether the student’s input-sensitivity geometry differs between the distribution it was trained on and the distribution it is evaluated against.

We use a coordinate-level participation-ratio measure

$$d_{\text{coord}} = \frac{\left(\sum_j I_j\right)^2}{\sum_j I_j^2}, \quad I_j(\pi) = \frac{1}{n} \sum_i \|J_i[:, j]\|_1,$$

which captures how broadly the student’s per-coordinate sensitivity is distributed. This is a coordinate-level quantity and is distinct from the eigenvalue-based d_{sens} used in Appendix D.7; the two are not directly comparable.

The sign of $\Delta = d_{\text{coord}}^{\text{synth}} - d_{\text{coord}}^{\text{real}}$ depends on observation dimensionality. In Hopper-v5 ($d_{\text{obs}} = 11$), Δ is small and positive: student sensitivity is slightly more diffuse on synthetic inputs, but the magnitude is below seed-level noise. In Ant-v5 ($d_{\text{obs}} = 105$), Δ is positive and modest (+1.7 to +6.3), consistent with iid synthetic samples covering more input coordinates than correlated trajectory observations. In Humanoid-v5 ($d_{\text{obs}} = 348$), the pattern reverses and Δ is strongly negative: student sensitivity is substantially more concentrated on synthetic inputs ($\Delta \approx -23$ to -33). This dimension-dependent reversal is a direct prediction of concentration of measure in high dimensions, where iid samples in the ambient cube project tightly onto any fixed eigendirection of G_π . Appendix D.13 examines this mechanism directly.

D.12 Trajectory-control statistical analysis

This section reports the per-budget paired comparison between Variant A (contiguous trajectory chunks) and Variant B (iid samples from a buffer of teacher rollouts) used in Section 5.3 of the main

Table 35: Decomposition of recovery improvement from $\mathcal{N}(0, I)$ to $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$. Centering effect is $\mathcal{N}(\hat{\mu}, I) - \mathcal{N}(0, I)$; covariance effect is $\mathcal{N}(0, \hat{\Sigma}) - \mathcal{N}(0, I)$; interaction is the residual. Shares are percentages of the total gap.

Environment	Centering (pp)	Covariance (pp)	Interaction (pp)	Total (pp)
Humanoid-v5	+6.7 (13%)	+34.8 (66%)	+11.4 (21%)	+52.8
Ant-v5	+5.0 (24%)	+18.9 (89%)	-2.8 (-13%)	+21.1

Table 36: Hopper-v5 student d_{coord} on synthetic vs. real inputs. Mean $\pm$ std across 10 seeds.

Distribution	d_{coord} (synth.)	d_{coord} (real)	Δ (synth. - real)
Gaussian, $\sigma = 1$	9.8 ± 0.5	9.4 ± 0.7	+0.3
Gaussian, $\mu = 1, \sigma = 1$	9.8 ± 0.8	9.3 ± 0.7	+0.5
Uniform $[-1, 1]$	9.5 ± 0.5	9.4 ± 0.7	+0.1

Table 37: Ant-v5 student d_{coord} on synthetic vs. real inputs. Mean $\pm$ std across 10 seeds.

Distribution	d_{coord} (synth.)	d_{coord} (real)	Δ (synth. - real)
Gaussian, $\sigma = 1$	80.9 ± 2.6	76.8 ± 4.2	+4.2
Gaussian, $\mu = 1, \sigma = 1$	84.4 ± 2.8	78.1 ± 3.6	+6.3
Uniform $[-1, 1]$	79.0 ± 2.7	77.3 ± 4.5	+1.7

Table 38: Humanoid-v5 student d_{coord} on synthetic vs. real inputs. Mean $\pm$ std across 10 seeds.

Distribution	d_{coord} (synth.)	d_{coord} (real)	Δ (synth. - real)
Gaussian, $\sigma = 1$	310.6 ± 2.7	338.5 ± 1.5	-27.9
Gaussian, $\mu = 1, \sigma = 1$	315.7 ± 2.5	338.8 ± 1.5	-23.2
Uniform $[-1, 1]$	303.2 ± 3.1	336.0 ± 1.7	-32.8

text. The paired difference $\Delta = B - A$ is computed for each (environment, budget) pair across 10 matched student seeds. Both a paired two-sided t -test and a binomial sign test are reported.

Table 39: Hopper-v5 trajectory-control comparison. Mean $\pm$ SEM, 10 seeds per condition.

Budget (k)	Variant A (%)	Variant B (%)	Δ	$n(B > A)$	p_t	p_{sign}
0	94.1 ± 14.4	94.9 ± 13.8	$+0.8 \pm 3.0$	8/10	0.811	0.109
5	96.8 ± 12.5	97.5 ± 14.4	$+0.7 \pm 4.1$	2/10	0.874	0.109
10	95.8 ± 13.0	99.2 ± 14.2	$+3.4 \pm 2.9$	5/10	0.299	1.000
20	96.9 ± 14.6	97.4 ± 14.8	$+0.5 \pm 1.9$	5/10	0.791	1.000
50	97.1 ± 14.2	98.3 ± 13.9	$+1.2 \pm 1.8$	4/10	0.542	0.754
100	100.0 ± 14.1	99.4 ± 14.5	-0.6 ± 1.4	4/10	0.694	0.754

Table 40: Walker2d-v5 trajectory-control comparison. Mean $\pm$ SEM, 10 seeds per condition.

Budget (k)	Variant A (%)	Variant B (%)	Δ	$n(B > A)$	p_t	p_{sign}
0	81.0 ± 11.9	81.0 ± 11.9	$+0.0 \pm 0.0$	—	—	—
5	93.6 ± 11.2	90.6 ± 12.1	-3.0 ± 5.6	5/10	0.609	1.000
10	93.7 ± 9.9	94.4 ± 10.9	$+0.8 \pm 5.0$	4/10	0.878	0.754
20	95.7 ± 12.5	91.5 ± 13.6	-4.1 ± 7.4	4/10	0.592	0.754
50	93.2 ± 15.0	96.5 ± 12.8	$+3.3 \pm 6.8$	5/10	0.639	1.000
100	89.9 ± 11.0	92.6 ± 11.3	$+2.7 \pm 3.8$	4/10	0.503	0.754

No (environment, budget) combination reaches significance at $\alpha = 0.05$ on either test. Sign-test counts cluster around 5/10 across all conditions, consistent with the null hypothesis of no systematic preference. With $n = 10$ seeds and an empirical residual standard deviation of approximately $\sigma \approx 4$ percentage points, the paired test has $\sim 80\%$ power to detect differences of 2–3 percentage points

Table 41: Ant-v5 trajectory-control comparison. Mean $\pm$ SEM, 10 seeds per condition.

Budget (k)	Variant A (%)	Variant B (%)	Δ	$n(B > A)$	p_t	p_{sign}
0	88.9 $\pm$ 2.7	88.1 $\pm$ 3.0	-0.8 $\pm$ 2.6	5/10	0.774	1.000
5	97.8 $\pm$ 2.2	98.8 $\pm$ 2.2	+1.0 $\pm$ 1.3	6/10	0.471	0.754
10	98.5 $\pm$ 2.0	98.9 $\pm$ 2.2	+0.4 $\pm$ 1.6	4/10	0.831	0.754
20	98.7 $\pm$ 2.1	99.3 $\pm$ 2.4	+0.5 $\pm$ 1.8	6/10	0.788	0.754
50	100.4 $\pm$ 1.9	96.4 $\pm$ 2.5	-4.0 $\pm$ 1.8	3/10	0.064	0.344
100	100.0 $\pm$ 1.9	99.4 $\pm$ 2.3	-0.6 $\pm$ 1.8	5/10	0.758	1.000

Table 42: Humanoid-v5 trajectory-control comparison. Mean $\pm$ SEM, 10 seeds per condition.

Budget (k)	Variant A (%)	Variant B (%)	Δ	$n(B > A)$	p_t	p_{sign}
0	35.4 $\pm$ 6.5	37.4 $\pm$ 6.7	+2.0 $\pm$ 1.1	8/10	0.124	0.109
5	80.1 $\pm$ 5.8	81.4 $\pm$ 6.6	+1.3 $\pm$ 2.9	6/10	0.685	0.754
10	86.3 $\pm$ 5.4	91.1 $\pm$ 3.9	+4.8 $\pm$ 4.1	5/10	0.287	1.000
20	93.6 $\pm$ 3.8	92.1 $\pm$ 4.5	-1.5 $\pm$ 2.3	4/10	0.543	0.754
50	98.5 $\pm$ 3.4	95.4 $\pm$ 4.0	-3.0 $\pm$ 1.8	5/10	0.142	1.000
100	100.0 $\pm$ 3.8	102.4 $\pm$ 3.0	+2.4 $\pm$ 2.4	5/10	0.367	1.000

(two-sided $\alpha = 0.05$). The observed effect sizes lie below this detection threshold, indicating that a substantially larger study would be needed to identify any small residual trajectory-structure effect.

D.13 Concentration-of-measure analysis for uniform-wide sampling

A consistent finding in our experiments is that students trained on Uniform $[-1, 1]$ synthetic observations achieve recovery rates competitive with Gaussian and Laplace baselines, even though uniform-wide samples are far from the empirical state distribution in ambient distance. This subsection provides a geometric explanation in terms of concentration of measure.

Setup. We compute the teacher Gram matrix $G_\pi = \mathbb{E}_{x \sim \text{real}}[J(x)^\top J(x)] \in \mathbb{R}^{348 \times 348}$ for the Humanoid-v5 teacher, where $J(x)$ is the Jacobian of the teacher’s deterministic action mean. G_π is estimated from 120 real teacher-rollout observations using finite differences ($\epsilon = 10^{-4}$). We project two sets of observations onto the top four eigenvectors of G_π : 400 real teacher-rollout observations, and 800 samples from the synthetic distribution Uniform $[-1, 1]^{348}$. The top four eigenvectors capture 31.5% of the total Jacobian variance and are chosen for visual clarity; the full sensitive subspace, as reported in Appendix D.7, has $d_{\text{sens}} = 15.5$ and the top 78 eigendirections capture 90% of the variance.

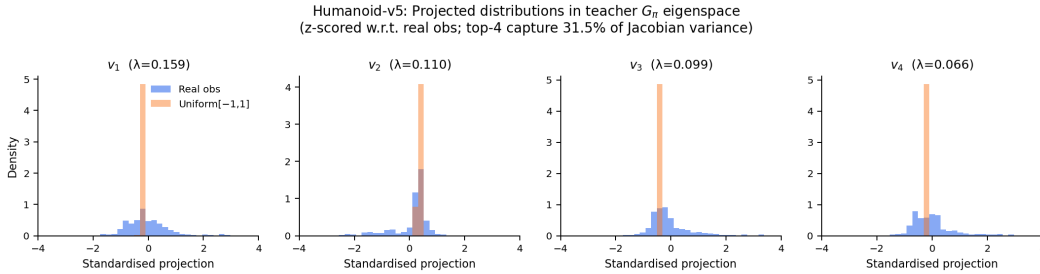


Figure 4: Projections of real (blue) and uniform-wide synthetic (orange) observations onto the top four eigenvectors of the Humanoid-v5 teacher Gram matrix. Real observations spread broadly along each eigendirection; synthetic observations concentrate sharply near zero, in agreement with concentration of measure.

Discussion. Synthetic projections concentrate sharply near zero along every sensitive eigendirection, while real observations spread broadly. This is the Johnson–Lindenstrauss prediction: a Uniform $[-1, 1]$ vector in $d = 348$ dimensions projected onto any fixed unit vector concentrates near

Table 43: Mean and standard deviation of the projection of real and synthetic Humanoid observations onto the top four eigenvectors of G_π .

	λ (Jacobian var.)	Real (mean $\pm$ std)	Synthetic (mean $\pm$ std)
v_1	0.159	$+9.52 \pm 35.71$	-0.002 ± 0.585
v_2	0.110	-16.45 ± 51.42	-0.015 ± 0.577
v_3	0.099	$+12.94 \pm 28.53$	$+0.001 \pm 0.585$
v_4	0.066	$+9.81 \pm 44.15$	$+0.026 \pm 0.570$

zero with standard deviation $1/\sqrt{3d} \approx 0.577$, an order of magnitude smaller than the spread of real observations along the same direction.

The success of uniform-wide sampling despite this concentration is consistent with the smoothness story of Section 4 of the main text. Because the teacher is locally Lipschitz on the policy-relevant region (Humanoid teacher Lipschitz ≈ 0.25 , Appendix D.6), accurate imitation on a dense cluster of synthetic points near the real manifold suffices for deployment-time performance. The student does not need to be trained at points far from the real manifold; concentration of measure ensures that uniform-wide synthetic samples cluster near the origin of the sensitive eigenspace, where the real observation distribution also lives.

In our experiments, distributions that satisfy two conditions—approximate centering on the real-observation mean and sufficient spread along the teacher’s sensitive eigendirections—produce similar recovery rates. Gaussian $\sigma = 1$, Laplace $b = 1$, and Uniform $[-1, 1]$ all satisfy these conditions for the environments studied (Appendix D.6). This is consistent with the dimension-dependent reversal of Δ in Appendix D.11: in high dimensions, synthetic inputs project onto a smaller effective subspace than real ones, but this does not impede recovery because the projection still covers the policy-relevant directions.

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